Exogenous Features in Intraday Forecasts of Closing-Half-Hour Variance

Bullet-point draft of the full paper

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Abstract

Do exogenous features improve an intraday forecast of the realized variance of the S&P 500 over the closing half hour, 15:30–16:00 ET? The forecast is issued at 15:30. Models are estimated on every half-hour interval and scored on that closing interval, from 3 January 2020 to 30 April 2024 ($T_{\text{oos}} = 1{,}082$). The benchmark predictor set is a HAR regression. The full set adds the exogenous series. On QLIKE the full set lowers average loss in every intraday model class, 4 of the 6 reductions significant at the 5% level. Squared error does not confirm the reductions. Every intraday forecast remains less accurate, on QLIKE, than a daily HAR regression fit on the closing interval alone, and the exogenous features narrow that gap. On days with a same-day SPX expiration, a sign strategy in a straddle has a higher Sharpe ratio with the full set in every intraday class.

Notes. Colored notes are draft comments, not paper text.
Red: two live statements disagree.
Orange: a sentence does not match the computation behind a printed number.
Blue: a reader cannot recover the input or the rule.
The first line is the decision. The second line is the problem. The rest is the exposition.

1 Introduction

- **Question.** Do exogenous features improve an intraday forecast of the realized variance of the S&P 500 over the closing half hour?
  - At 15:30 ET on day t the forecast is of RV_t, the realized variance from 15:30 to 16:00 (Section 3).
  - The models are estimated on every 30-minute interval, $M = 48$, and the forecast that is scored is the one issued at 15:30.
  - The comparison is the full predictor set against the HAR benchmark. The exogenous series are the difference between the two (Section 4).
- **Why the HAR benchmark is the baseline.** A HAR regression forecasts variance from its own lags (Corsi, 2009). It does not use the other series observed by 15:30: realized moments, liquidity, the cross-section of returns, sentiment, implied volatility, volatility demand, and scheduled announcements.
- **Design.** One forecast function, for each intraday interval (t, i),

$$\hat{y}_{t,i} = f_{m,\mathcal{X},\mathcal{I}}(\boldsymbol{x}_{t,i}),$$

with model class m and predictor set $\mathcal{X} \in \{\text{HAR}, \text{Full}\}$.

  - The results in Section 6 use the intraday scheme, $\mathcal{I} = \{1, \ldots, M\}$.
  - The benchmark forecast is the daily OLS on the HAR benchmark, estimated on the closing interval only (Sections 5.2 and 5.3).
  - The loss is average QLIKE on the evaluation span $\mathcal{T}_{\text{oos}}$, 3 January 2020 to 30 April 2024, $T_{\text{oos}} = 1{,}082$ (Section 5.3).
- **Findings.**
  - The full predictor set lowers average QLIKE in every intraday model class (6 classes, 4 of the reductions significant at the 5% level; Section 6). The tree ensembles have the largest reductions. Among the linear models the elastic net's reduction is significant.
  - Squared error does not confirm the reductions: the full set raises the MSE in 3 of the 6 classes, and no MSE difference is significant.
  - Every intraday forecast has a higher average QLIKE than the benchmark forecast. For the linear models the difference is not significant at the 5% level; for the trees it is. The full predictor set narrows that gap in every model class (Section 6.4).
- **Economic application.** On days with a same-day SPX expiration, the sign of the variance forecast minus implied variance is a position in a zero-days-to-expiration straddle (Section 7). The full predictor set raises that strategy's Sharpe ratio in every intraday class (Section 7.3).
- **Roadmap.** Related work (Section 2), the data (Section 3), the predictors (Section 4), the methods (Section 5), the experiments (Section 6), the economic value (Section 7) and the conclusion (Section 8).

2 Related work

- **Realized variance and the HAR model:** $\mathbb{E}_t[\mathrm{RV}_{t+1}^{(d)}] = \beta_0 + \beta_d \mathrm{RV}_t^{(d)} + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)}$, with $\mathbb{E}_t$ the expectation given the information at the forecast origin on day t and $\mathrm{RV}_t^{(d)}$, $\mathrm{RV}_t^{(w)}$ and $\mathrm{RV}_t^{(m)}$ the average daily realized variance over the last 1, 5 and 22 trading days. The components are daily realized variance at the daily, weekly, and monthly horizons. An intraday interval as the object of the forecast is outside this regression.
  Corsi (2009), Andersen et al. (2003), Barndorff-Nielsen and Shephard (2004), Andersen et al. (2007), Corsi et al. (2008), Bollerslev et al. (2016), Bekierman and Manner (2018), Bennedsen et al. (2022), Hansen et al. (2012), Andersen et al. (2006).
  - Panel HAR estimates one regression jointly across assets: Bollerslev et al. (2018). Its sample is a cross-section of names.
  - Barndorff-Nielsen and Shephard (2002) (realized variance).
- **Diurnal pattern:** volatility follows a U-shaped intraday pattern. The standard adjustment divides intraday variance by a factor estimated from the same clock time on earlier days.
  Andersen and Bollerslev (1997) (intraday periodicity), Engle and Gallo (2006) (intraday measures as indicators of daily volatility), Hizmeri and Bevilacqua (2025) (morning volatility uncertainty).
  - Taylor and Xu (1997) (diurnal adjustment), Engle and Sokalska (2012) (intraday volatility).
- **0DTE options and the variance risk premium:** the variance risk premium, implied minus expected realized variance, $\mathbb{E}_t^{\mathbb{Q}}[\mathrm{RV}_t] - \mathbb{E}_t^{\mathbb{P}}[\mathrm{RV}_t]$ under the risk-neutral ($\mathbb{Q}$) and physical ($\mathbb{P}$) measures, with RV_t the realized variance to expiration, is positive on average.
  Hizmeri and Bevilacqua (2025); option pricing and implied volatility Black and Scholes (1973),

Merton (1973).

  - 0DTE options: Bandi et al. (2023), Beckmeyer et al. (2023), Brogaard et al. (2023), Dim et al. (2024), Vilkov (2026), Almeida et al. (2025); the variance risk premium: Carr and Wu (2009), Bollerslev et al. (2009).

- **Machine learning for volatility and returns:** a nonlinear function $f(\boldsymbol{x}_t)$ of a vector $\boldsymbol{x}_t$ of many predictors.
  Gu et al. (2020) (returns), Christensen et al. (2023) (realized volatility); gradient-boosted trees Ke et al. (2017) (LightGBM), Chen and Guestrin (2016) (XGBoost); hyperparameter optimization Akiba et al. (2019) (Optuna).
  - Breiman (2001) (random forests), Friedman (2001) (gradient boosting); realized volatility: Bucci (2020), Rahimikia and Poon (2020), Zhang et al. (2024).
- **Shrinkage with many weak signals:**

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda(\alpha\|\boldsymbol{\beta}\|_1 + \tfrac{1-\alpha}{2}\|\boldsymbol{\beta}\|_2^2)$$

  for a response vector $\boldsymbol{y}$, a predictor matrix $\boldsymbol{X}$ and a penalty strength λ (ridge $\alpha = 0$, lasso $\alpha = 1$); multi-penalty ridge gives each block of predictors its own penalty, $\lambda \sum_j \omega_j \beta_j^2$ with the penalty weight ω_j constant within a block.
  Giannone et al. (2021) (the illusion of sparsity), Shen and Xiu (2025) (weak signals), Kozak et al. (2020) (shrinking the cross-section), Hoerl and Kennard (1970) (ridge), Tibshirani (1996) (lasso), Zou and Hastie (2005) (elastic net), van de Wiel et al. (2021) (multi-penalty ridge), van Wieringen (2023).
  - Zou (2006) (adaptive lasso, penalty weights), Friedman et al. (2010) (elastic-net parametrization).

3 Data

- **Sources**. Eight groups. The closing-interval models use all 47 exogenous series. The intraday models use the 43 series in every group except FOMC. Appendix Table 5 defines each series.
  - **Realized moments** (7). Front-month E-mini S&P 500 futures, the most active contract, nearly 24 hours a day, 1998–2024, on a 30-minute grid. Trading day $t = 1, \ldots, T$; intraday interval $i = 1, \ldots, M$, the half hours of the day, each ending on a half-hour stamp; the closing interval $i = c$ is 15:30–16:00 ET. Each moment on (t, i) sums a one-minute quantity over the non-missing minutes of that half hour.
  - $r_{t,i,k}$: the one-minute change in the log price of that contract, the k-th non-missing minute in interval i of day t, $k = 1, \ldots, N_{t,i}$, with $N_{t,i} \leq 30$. Realized variance (Andersen et al., 2003):

$$\mathrm{RV}_{t,i} = \sum_{k=1}^{N_{t,i}} r_{t,i,k}^2.$$

  - The 7 exogenous moments, the same one-minute sums. Realized variance above is the target and is not one of them:

$$\sum_k r_{t,i,k}, \quad \sum_k |r_{t,i,k}|, \quad \sum_k r_{t,i,k}^3, \quad \sum_k r_{t,i,k}^4, \quad \mathrm{RS}_{t,i}^{+} = \sum_k r_{t,i,k}^2 \, \mathbf{1}\{r_{t,i,k} > 0\},$$

$$\mathrm{BV}_{t,i} = \sum_{k \geq 2} |r_{t,i,k}| \, |r_{t,i,k-1}|, \qquad \mathrm{AC}_{t,i} = \sum_{k \geq 2} r_{t,i,k} \, r_{t,i,k-1},$$

The third and fourth sums are the realized skewness and kurtosis of Amaya et al. (2015) without their normalizations (division by $\mathrm{RV}_{t,i}^{3/2}/\sqrt{N_{t,i}}$ and by $\mathrm{RV}_{t,i}^{2}/N_{t,i}$). $\mathrm{RS}_{t,i}^{+}$ is the positive realized semivariance (Barndorff-Nielsen et al., 2010); $\mathrm{BV}_{t,i}$ is the bipower variation of Barndorff-Nielsen and Shephard (2004) without its $\pi/2$ scaling; $\mathrm{AC}_{t,i}$ is the first-order realized autocovariance.

– **Liquidity** (14). On the same futures contract: volume, the number of front-month contracts traded in the half hour, and the minute count $N_{t,i}$. On S&P 500 constituents, 1993–2024, equal- and value-weighted: turnover, buyer- and seller-initiated turnover, the effective spread, the quoted spread, and order-flow imbalance. Turnover is shares traded in the half hour divided by the previous day's shares outstanding. Buyer- and seller-initiated turnover keep the estimated side of the trade. The effective spread is the mean percent difference between the trade price and the preceding midquote. The quoted spread is the mean percent difference between the bid and the ask. Equal weights are simple averages. Value weights are the previous day's market capitalization. Order-flow imbalance, for each weighting ($\mathrm{OFI}^{\mathrm{EW}}$, $\mathrm{OFI}^{\mathrm{VW}}$):

$$\mathrm{OFI}_{t,i} = \frac{\mathrm{Buy}_{t,i} - \mathrm{Sell}_{t,i}}{\mathrm{Buy}_{t,i} + \mathrm{Sell}_{t,i}} \in [-1, 1], \qquad \text{missing if } \mathrm{Buy}_{t,i} + \mathrm{Sell}_{t,i} = 0,$$

with $\mathrm{Buy}_{t,i}$ and $\mathrm{Sell}_{t,i}$ the buy and sell turnover in interval i of day t.

– **Equal-weighted market** (6) and **value-weighted market** (6). Six constituent moments under each weighting: realized variance, skewness, quarticity, absolute variation, bipower variation, and positive semivariance, with the weights above. These sums use 5-minute log returns. The futures moments use one-minute log-price changes. The constituent list has no signed return and no autocovariance. The futures list has no realized variance, which is the target.

– **Sentiment** (3). StockTwits messages, from 2009 (posts) and 2010 (tagged posts). Attention is the number of posts in the half hour. The sentiment score is 1 when every tagged post is bullish and −1 when every tagged post is bearish. The sentiment count is the number of posts that carry a bullish or bearish tag.

– **Implied volatility** (3). Cboe indices: VIX (from 2006), the 30-day S&P 500 index; VIX3M (from 2009), the three-month index; VVIX (from 2012), the implied volatility of the VIX. All end in February 2024.

– **Volatility demand** (4). Cboe option volume, 2011 to August 2023: net vega bought by non-market-makers. Two universes, options on the S&P 500 and the top S&P 500 exchange-traded funds, and all traded options. Two trade sets, trades that open or close a position, and trades that open a position.

– **FOMC** (4), closing-interval models only. Scheduled announcement times, 1993 to November 2023. The four series are the announcement-interval indicator, the announcement-day indicator, and $1/(1 + n_{t,i}^{+})$ and $1/(1 + n_{t,i}^{-})$, with $n_{t,i}^{+}$ and $n_{t,i}^{-}$ the number of intervals to the next and since the last scheduled announcement. The schedule is known before the interval.

– *[name the vendor of each file].*

E2. Decision: name the vendor, the price field, the roll, and the bin edges.
Problem: the vendor, the price inside the one-minute log change, the day whose volume picks the front month, and the clock edges of every bin except 15:30–16:00 and the weekend break are missing, so the panel cannot be rebuilt.
The futures moments are now sums of one-minute changes in the log price of the front-month contract, the most active. The text does not say whether that price is a last trade, a midpoint, or a settlement, and it does not name the vendor of the file. It does not say which day's volume makes a contract the most active. The closing bin is 15:30–16:00 ET, and the grid drops Friday 20:00 through Sunday 18:30. Each other bin is a half hour ending on a half-hour stamp. Those stamps are not listed, so the $M = 48$ bins cannot be rebuilt from the text.

- **Intraday grid and cleaning.**
  - Grid: 30-minute intervals from the first observed realized variance. Interval i of day t is retained if and only if its realized variance is observed and it lies outside the weekend break:
$$(t, i) \in \mathcal{G} \iff \mathrm{RV}_{t,i} \text{ observed and } (t, i) \notin (\text{Fri } 20{:}00,\ \text{Sun } 18{:}30),$$
    with $\mathcal{G}$ the interval grid, the set of retained intervals.
  - Lags run backward through the retained intervals in calendar order, across days and overnight: $(t, i-j)$ denotes the j-th retained interval before (t, i), possibly on an earlier trading day.
  - Exchange holidays and early-close days remain in the futures sample; the option-trading sample excludes early-close days.
  - Exogenous series u: availability (missing-data) indicator, recorded before any filling,
$$a_{t,i} = \mathbf{1}\{u \text{ observed in interval } (t, i)\} \in \{0, 1\}.$$
  - Then $u_{t,i}$ is forward-filled (last observation carried forward), without a limit on staleness. Every interval with $a_{t,i} = 0$ is later replaced by the imputation fill of Section 4, step 4.
  - The diurnal adjustment $s^2_{t,i}$ and the winsorized target $\tilde{y}_{t,i}$ are defined in Section 5.1. Window lengths and quantile levels: Appendix B.1.
- **Samples** (Appendix Table 3).
  - Intraday sample: 300,317 half-hour intervals from June 1998, with rolling estimation windows of $W = 24{,}000$ intervals (trees) and $W = 96{,}000$ intervals (linear models; Appendix B.2):
$$300{,}317 = 303{,}442 \text{ intervals on the grid} - 3{,}125 \text{ burn-in intervals.}$$
  - Closing-interval sample ($i = c$): 3,469 trading days, July 2010 to April 2024, with a rolling estimation window of $W = 2{,}000$ trading days.
  - 1,469 rolling-window forecasts from June 2018:
$$1{,}469 = 3{,}469 - 2{,}000.$$
  - Hyperparameter selection for the linear models on a pre-sample: closing-interval observations from 13 April 2004; validation samples from 6 January 2016 to 2 January 2020.
  - Evaluation span $\mathcal{T}_{\text{oos}}$: 3 January 2020 to 30 April 2024, the $T_{\text{oos}} = 1{,}082$ trading days on which every rolling forecast exists, less 6 early-close days. It starts with the intraday linear forecasts, after the pre-sample above (Section 5.3).
  - Option-data subsample $\mathcal{T}_{\text{opt}} \subset \mathcal{T}_{\text{oos}}$, on which the straddle strategy is evaluated (Section 7):
$$T_{\text{opt}} = 866 \text{ trading days} = 872 \text{ SPXW expirations} - 6 \text{ early-close days.}$$
  - The 419 later expirations (413 full trading days) are held out.

4 Predictors

- **Eight groups of exogenous predictors, 47 series:** realized moments (7), liquidity (14), equal-weighted market (6), value-weighted market (6), sentiment (3), implied volatility (3), volatility demand (4), FOMC (4). Definitions: Section 3. Each series: Appendix Table 5.

$$47 = 7 + 14 + 6 + 6 + 3 + 3 + 4 + 4.$$

  - FOMC:

$$\mathbf{1}\{\text{announcement in } (t,i)\}, \quad \mathbf{1}\{\text{announcement on day } t\}, \quad 1/(1+n^{+}_{t,i}), \quad 1/(1+n^{-}_{t,i}).$$

  - $n^{\pm}_{t,i}$: number of intervals to the next (+) and since the last (−) scheduled announcement.

- **Predictors built from each series.**
  - Trailing averages of the transformed series $\tilde{u}$ over horizons of k intervals, lagged one interval (HAR components; Corsi, 2009):

$$\bar{u}^{(k)}_{t,i} = \frac{1}{k}\sum_{j=1}^{k} \tilde{u}_{t,i-j}, \qquad k \in \mathcal{K},$$

$$\mathcal{K} = \begin{cases} \{1, 5, 25, 125, 625, 3125\} & \text{closing-interval model,} \\ \{2^0, 2^1, \ldots, 2^{11}\} \text{ (base 2, 12 horizons)} & \text{intraday model.} \end{cases}$$

  - k counts intervals of the full intraday grid, overnight included; the closing-interval model uses the rows $i = c$.
  - The same averages of the availability indicator, $\bar{a}^{(k)}_{t,i}$.
  - Averages of an activity indicator, $\bar{\chi}^{(k)}_{t,i}$, for 7 zero-inflated series.

    E6. Decision: one rule and one count.
    Problem: the bullet uses 7 zero-inflated series, the transformation uses the 7 series that pass a 20% mode rule, and the intraday column count uses 5. Those are three different sets.
    This bullet says the activity average $\bar{\chi}^{(k)}_{t,i}$ is built for 7 series, and it calls those series zero-inflated. That word means the raw series has a pile of exact zeros. The transformation further down does not look for zeros. It sets the indicator $\chi_{t,i}$ to 1 when the value differs from the series' full-sample mode, and it keeps a series only when that mode covers at least 20% of intervals. A mode can sit at a number other than zero, so the 20% rule and a mass at zero select different series. The number of series that pass the 20% rule is 7. The intraday column formula then multiplies by a third count, 5. The activity block of the design is the product of one of these counts with the horizons, and the three counts are not the same count.

  - Names in tables and figures: the bar and the interval index are dropped ($i = c$ in the closing-interval model), so $u^{(k)}_t$ is series u averaged over k intervals; e.g. $\text{VIX}^{(25)}_t$ is the transformed VIX averaged over the 25 intervals ending at 15:30 on day t. Series symbols: VIX, VIX3M, VVIX; constituent aggregates carry EW or VW, as in OFI^{EW}, RV^{VW}, $\text{Spread}^{\text{VW}}$; FOMC. Indicator averages: $\bar{a}^{(k)}_t(u)$, $\bar{\chi}^{(k)}_t(u)$.

- **HAR benchmark:** the HAR components of the target, $\bar{y}^{(k)}_{t,i}$, the same average as $\bar{u}^{(k)}_{t,i}$ applied to the winsorized target (6 columns), and the calendar columns $\boldsymbol{D}_{t,i}$: the five weekday dummies, the hour of the bar-end stamp, and the overnight indicator.

E9. Decision: the calendar stops at the overnight indicator, and that indicator stays only if the overnight intervals stay in the training sample (note E15).
Problem: the columns after the overnight indicator are out of the specification, and the printed column counts and Table 1 still include them. Whether the overnight indicator itself stays depends on the sample in note E15.
The sentence above keeps three calendar pieces: the five weekday dummies, the hour of the bar-end stamp, and the overnight indicator. Every column that used to sit after the overnight indicator is out of the specification. That is the opening-interval indicator, the closing-interval indicator, expiration, expiration week, quadruple witching, the quadruple-witching close, month end, quarter end, days to expiration, and the products of the HAR components with the opening and closing indicators. The overnight indicator's own place depends on which sample note E15 keeps. On all 48 intervals it equals 1 off the cash session and again on the 16:00 bar, so it varies across the window and a regression can estimate it. On the 13 regular-hours bars it equals 1 only on the 16:00 bar, so inside that sample it is constant on every bar but the close. The column counts in the formulas below, and the fits in Table 1, were built with the dropped columns still in the matrix.

- **Transformations, in order:**
  1. diurnal adjustment, dividing by a factor estimated on the same intraday interval ($\mathcal{D}_{t,i}$ as for $s^2_{t,i}$) and bounded relative to its own trailing median:

$$\tilde{u}_{t,i} = \frac{u_{t,i}}{b^{(u)}_{t,i}}, \qquad b^{(u)}_{t,i} = \begin{cases} \operatorname{clip}\big(\operatorname{mean}_{t' \in \mathcal{D}_{t,i}} u_{t',i},\ 0.5\,\rho_{t,i},\ 2\rho_{t,i}\big), & u \geq 0, \\ \max\big\{\operatorname{clip}\big(\operatorname{sd}_{t' \in \mathcal{D}_{t,i}} u_{t',i},\ 0.5\,\rho_{t,i},\ 2\rho_{t,i}\big),\ 0.1\,\bar{\sigma}^{(u)}\big\}, & u \text{ signed}; \end{cases}$$

     - $\rho_{t,i}$: median of the 250 most recent unbounded factors of interval i (at least 20 required).
     - $\bar{\sigma}^{(u)}$: full-sample median of the standard deviation on the same intraday interval.
     - Factor of a nonnegative series zero or undefined: $b^{(u)}_{t,i} = 1$.
     - Not adjusted: VIX, VIX3M and VVIX, StockTwits sentiment, FOMC.
  2. a variance-stabilizing transformation by series type, $\tilde{u}_{t,i} \leftarrow g(\tilde{u}_{t,i})$, the first matching rule applying:

$$g(v) = \begin{cases} \sqrt{v} & \text{squared returns, realized variance, turnover,} \\ & \text{bipower variation, effective spread,} \\ \operatorname{sign}(v)\,|v|^{1/2} & \text{autocovariance,} \\ v^{1/3} & \text{cubed returns,} \\ \operatorname{sign}(v)\,|v|^{1/4} & \text{fourth-power returns,} \\ v & \text{other signed series, absolute returns,} \\ \log \max(v, 10^{-12}) & \text{otherwise.} \end{cases}$$

     - Signed series with no type rule whose standard-deviation divisor sits at its floor on at least 50% of intervals (volatility demand) instead take an inverse hyperbolic sine of the raw series:

$$\tilde{u}_{t,i} = \operatorname{asinh}(u_{t,i}/\kappa_{t,i}), \qquad \kappa_{t,i} = \max\big\{e_{t,i-1},\ 0.01\,\operatorname{med}_{j \geq 1}\, e_{t,i-j}\big\}.$$

     - $e_{t,i}$: exponentially weighted moving average of $|u|$ over intervals with $u \neq 0$, half-life 250 days.
  3. winsorization at the same rolling quantiles as the target (Section 5.1), on the series' own preceding intervals;

4. imputation of a missing value by the median of the series' observed values on earlier intervals, kept alongside the availability indicator:

$$\tilde{u}_{t,i} \leftarrow \begin{cases} \tilde{u}_{t,i}, & a_{t,i} = 1, \\ \operatorname{med} \mathcal{A}_{t,i}, & a_{t,i} = 0,\ \mathcal{A}_{t,i} \neq \emptyset, \\ \tilde{u}_{(1)}, & a_{t,i} = 0,\ \mathcal{A}_{t,i} = \emptyset, \end{cases}$$

   - $\mathcal{A}_{t,i} = \{\tilde{u}_{t',i'} : (t',i') \prec (t,i),\ a_{t',i'} = 1\}$, where $(t',i') \prec (t,i)$ means (t',i') is a retained interval earlier than (t,i).
   - $\tilde{u}_{(1)}$: the adjusted value on the series' first observed interval.
5. activity indicators, for the 7 series whose full-sample mode accounts for at least 20% of intervals:

$$\chi_{t,i} = \mathbf{1}\{u_{t,i} \neq \operatorname{mode}(u)\};$$

6. trailing averages $\bar{u}^{(k)}_{t,i}$, $\bar{a}^{(k)}_{t,i}$, $\bar{\chi}^{(k)}_{t,i}$, $\bar{y}^{(k)}_{t,i}$;
7. robust standardization on the rolling estimation window (median and interquartile range, with floors), for the averages of series values only, under the intraday scheme. The median and the interquartile range use only intervals with $a_{t',i'} = 1$ inside $\mathcal{W}_{t,i}$ (Appendix B.2). They are evaluated every 480 intervals and held until the next point. $M^{(k)}_{t,i}$ is the largest such interquartile range at a grid point up to (t,i). The 0.01 floor applies. $\mathrm{IQR}^{*(k)}$ does not enter this denominator.

$$z^{(k)}_{t,i} = \begin{cases} \dfrac{\bar{u}^{(k)}_{t,i} - \operatorname{med}\{\bar{u}^{(k)}_{t',i'} : (t',i') \in \mathcal{W}_{t,i},\ a_{t',i'} = 1\}}{\max\{\mathrm{IQR}^{(k)}_{t,i},\ 0.01\, M^{(k)}_{t,i}\}} & \text{if at least 512 such intervals,} \\ 0 & \text{otherwise,} \end{cases}$$

$$\mathrm{IQR}^{(k)}_{t,i} = \mathrm{IQR}\{\bar{u}^{(k)}_{t',i'} : (t',i') \in \mathcal{W}_{t,i},\ a_{t',i'} = 1\}.$$

   - On the first window the two statistics use that window's own rows. An interquartile range below 10^{-12} is set to 1 there and, at a later grid point, leaves the previous scale in place.
   - $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$: not standardized.
   - HAR components and calendar dummies are not on this denominator. Their median and interquartile range use every row of $\mathcal{W}_{t,i}$, with $\mathrm{IQR}^{*(k)} = 0$, and the same 0.01 floor against the largest interquartile range of that column so far.

- **Column screening:** at every estimation, on the window $\mathcal{W}_t$, keep

$$\mathcal{J}_t = \{j :\ \boldsymbol{x}_{\cdot j} \text{ not constant},\ \boldsymbol{x}_{\cdot j} \neq \boldsymbol{x}_{\cdot j'}\ \forall\, j' < j,\ j' \in \mathcal{J}_t\}, \qquad \boldsymbol{x}_{\cdot j} = (x_{t'j})_{t' \in \mathcal{W}_t}.$$

  - Equality is exact (bitwise); the first copy is kept.
  - The closing-interval model with the full predictor set keeps a median of 369 of its 628 columns.

- **Predictor sets:** the HAR benchmark has 22 columns; the full predictor set has 628.

$$628 = \underbrace{22}_{\bar{y}^{(k)},\ \boldsymbol{D}} + \underbrace{47 \times 6}_{\bar{u}^{(k)}} + \underbrace{47 \times 6}_{\bar{a}^{(k)}} + \underbrace{7 \times 6}_{\bar{\chi}^{(k)}}, \qquad 22 = 6 + 16.$$

– The intraday model's full predictor set has 1,144 columns: 43 series, 12 horizons, and 24 interactions of the HAR components with the opening- and closing-interval dummies. Those counts, and 22, still include the calendar columns and products dropped above (note E9).

$$1{,}144 = \underbrace{12 + 16}_{\bar{y}^{(k)},\ \boldsymbol{D}} + \underbrace{24}_{\bar{y}^{(k)} \times \text{open, close}} + 43 \times 12 + 43 \times 12 + 5 \times 12.$$

The 43 intraday series are every group in Section 3 except FOMC. The series left out of that matrix are the FOMC group, 4 series. Appendix Table 5 names each series.

- **No look-ahead:** every statistic dated (t, i) uses only intervals before (t, i), so every closing-interval predictor is known at the 15:30 ET forecast origin: the trailing averages, $s^2_{t,i}$, the winsorization bounds, $\rho_{t,i}$, the standardization statistics and $\mathcal{J}_t$.
  - Full-sample elements, disclosed: the mode, $\bar{\sigma}^{(u)}$ and the assignment of series to the inverse hyperbolic sine.
  - The FOMC distances use the scheduled announcement calendar, which is known before the interval (Section 3).
  - An unobserved interval before the series' first print takes that first observed value (step 4).
  - The first window's standardization statistics use warm-up rows only.

5 Methods

5.1 Problem formulation

- **Rolling-window out-of-sample exercise,** re-estimated every period; every predictor is in the information set $\mathcal{F}_t$ at the 15:30 ET forecast origin. A period is a 30-minute intraday interval under the intraday scheme (Appendix B.2) and a trading day for the benchmark forecast (Section 5.3).
- **Notation:** $t = 1, \ldots, T$ indexes trading days and $i = 1, \ldots, M$ the 30-minute intraday intervals of day t, overnight included; $i = c$ is the closing interval, the closing half hour (15:30–16:00 ET).
  - Realized variance $\mathrm{RV}_{t,i}$ and the lag $(t, i-j)$: Section 3. The forecast target is the closing interval, $\mathrm{RV}_t \equiv \mathrm{RV}_{t,c}$.
  - $\mathcal{F}_t$: all data through 15:30 ET on day t; $\mathbb{E}_t[\cdot] \equiv \mathbb{E}[\cdot \mid \mathcal{F}_t]$ and $\mathrm{Var}_t(\cdot) \equiv \mathrm{Var}(\cdot \mid \mathcal{F}_t)$.
- **Diurnal adjustment** (Andersen and Bollerslev, 1997): the trailing mean of RV in the same interval, strictly before day t,

$$s^2_{t,i} = \frac{1}{|\mathcal{D}_{t,i}|} \sum_{t' \in \mathcal{D}_{t,i}} \mathrm{RV}_{t',i},$$

$$\mathcal{D}_{t,i} = \{\text{last } n_{\mathrm{d}} \text{ trading days } t' < t \text{ with interval } i\}, \qquad |\mathcal{D}_{t,i}| \geq \underline{n}_{\mathrm{d}}.$$

  - Intervals with $|\mathcal{D}_{t,i}| < \underline{n}_{\mathrm{d}}$: $s^2_{t,i} = 1$, and the interval is not scored. Values of n_{d} and $\underline{n}_{\mathrm{d}}$: Appendix B.1.
- **Target:** diurnally adjusted realized volatility (the square root stabilizes the variance), win-

sorized on that scale at rolling lower and upper empirical quantiles:

$$y_{t,i} = \sqrt{\mathrm{RV}_{t,i}/s^2_{t,i}}, \qquad \mathcal{Y}_{t,i} = \{y_{t,i-j} : j = 1, \ldots, n_{\mathrm{w}}\},$$
$$\tilde{y}_{t,i} = \mathrm{clip}(y_{t,i},\ \underline{Q}_{t,i},\ \overline{Q}_{t,i}).$$

  - $\underline{Q}_{t,i}$, $\overline{Q}_{t,i}$: lower and upper empirical quantiles of $\mathcal{Y}_{t,i}$, the previous n_{w} intervals, all intervals of the day pooled; clip truncates its first argument to the interval between the other two. Levels and n_{w}: Appendix B.1.
  - Models fit $\tilde{y}_{t,i}$; out-of-sample scores use the unwinsorized $\mathrm{RV}_{t,c}$.
  - $y_t \equiv y_{t,c}$ and $\tilde{y}_t \equiv \tilde{y}_{t,c}$: the closing interval's values.
- **Back-transform:** square and rescale, with no recalibration:

$$\hat{\sigma}^2_t = \hat{y}_t^{\,2}\, s^2_{t,c},$$

where $\hat{y}_t \equiv \hat{y}_{t,c}$ is the forecast of y_t made with $\mathcal{F}_t$.

E10. Decision: say $\hat{y}_t$ forecasts $\tilde{y}_t$.
Problem: the fit minimizes squared error in the clipped target $\tilde{y}$, and the back-transform sentence calls $\hat{y}_t$ a forecast of the unclipped y_t. QLIKE is then scored on the unwinsorized $\mathrm{RV}_{t,c}$.
y_t, defined as $y_{t,c}$, is the square root of realized variance on the closing interval before any clipping. $\tilde{y}_t$ is that square root after the rolling 1st and 99th percentiles have pulled the tails in, and the estimation objective chooses $\hat{y}$ by minimizing squared error in $\tilde{y}$, not in y. The back-transform undoes the diurnal adjustment by squaring the fitted value, $\hat{\sigma}^2_t = \hat{y}^2_t s^2_{t,c}$, and it sets $\hat{y}_t$ equal to the closing-interval fit $\hat{y}_{t,c}$. The sentence under the display then calls $\hat{y}_t$ a forecast of y_t. The loss that is scored, QLIKE, is computed from the unwinsorized $\mathrm{RV}_{t,c}$. The number that comes out of the regression is a fitted value for the clipped series. The sentence describes the square of that fitted value as a forecast of the unclipped series.

  - Jensen bias, downward, not corrected and disclosed: $\mathbb{E}_t[y_t^2] - (\mathbb{E}_t[y_t])^2 = \mathrm{Var}_t(y_t) \geq 0$.
- **Rolling estimation window.** Under the intraday scheme the window is $\mathcal{W}_{t,i}$ (Appendix B.2): the last W intervals before (t, i). $\mathcal{I}$ denotes the intervals used in estimation.
- **Hyperparameter-selection splits run forward in time** (Gu et al., 2020): an estimation sample, a gap of n_{gap} observations and a validation sample of n_{val} observations:

$$\mathcal{W}^{\mathrm{est}}_{t,i} = \{(t,\, i - W), \ldots, (t,\, i - n_{\mathrm{val}} - n_{\mathrm{gap}} - 1)\},$$
$$\mathcal{W}^{\mathrm{gap}}_{t,i} = \{(t,\, i - n_{\mathrm{val}} - n_{\mathrm{gap}}), \ldots, (t,\, i - n_{\mathrm{val}} - 1)\},$$
$$\mathcal{W}^{\mathrm{val}}_{t,i} = \{(t,\, i - n_{\mathrm{val}}), \ldots, (t,\, i - 1)\}.$$

  Interval (t, i) never enters. Values of n_{gap} and n_{val}: Appendix B.1.
- **Estimation objective:** squared error on the scale of $\tilde{y}$. The forecast of interval (t, i) is estimated on the full window:

$$\hat{\boldsymbol{\theta}}_{t,i} = \arg\min_{\boldsymbol{\theta}} \sum_{(t',i') \in \mathcal{W}_{t,i}} \left(\tilde{y}_{t',i'} - \hat{y}_{t',i'}(\boldsymbol{\theta})\right)^2.$$

  - (t', i') indexes the intervals of the window; $\boldsymbol{\theta}$: the model's parameters.
  - During hyperparameter selection only, each candidate configuration is estimated on the estimation sample $\mathcal{W}^{\mathrm{est}}_{t,i}$ in place of $\mathcal{W}_{t,i}$ and evaluated on the validation sample $\mathcal{W}^{\mathrm{val}}_{t,i}$.

5.2 Models

- **HAR and the HAR benchmark, by OLS** (Corsi, 2009), for each intraday interval (t, i):

$$\hat{y}_{t,i} = \beta_0 + \sum_{k \in \mathcal{K}} \beta^{(k)} \, \bar{y}_{t,i}^{(k)} + \boldsymbol{\delta}^\top \boldsymbol{D}_{t,i}.$$

  - $\bar{y}_{t,i}^{(k)}$ and $\mathcal{K}$: Section 4. $\bar{y}_{t,i}^{(k)}$ is the HAR component at horizon k, the mean of the winsorized target over the k half-hour intervals before (t, i), overnight included. The scored forecast is the closing interval, $\hat{y}_t \equiv \hat{y}_{t,c}$.
  - $\boldsymbol{D}_{t,i}$: calendar dummies, coefficients $\boldsymbol{\delta}$; HAR alone sets $\boldsymbol{\delta} = \mathbf{0}$. The HAR benchmark ($\mathcal{X} = \text{HAR}$) is the HAR components and the calendar dummies.
  - $\boldsymbol{x}_{t,i} \in \mathbb{R}^p$: the full predictor set ($\mathcal{X} = \text{Full}$), the HAR benchmark's predictors plus, for each additional series u, the HAR components $\bar{u}_{t,i}^{(k)}$, $k \in \mathcal{K}$, and the averages of its availability and activity indicators (Section 4).

- **Ridge (Hoerl and Kennard, 1970), lasso (Tibshirani, 1996) and elastic net (Zou and Hastie, 2005).** The penalty is added to the sum of squared residuals, as in the lasso of Zhang et al. (2024), and the coefficients are solved as the window rolls:

$$\hat{\boldsymbol{\beta}}_{t,i}^{\text{ridge}} = \underset{\beta_0, \boldsymbol{\beta}}{\arg\min} \sum_{(t',i') \in \mathcal{W}_{t,i}} (\tilde{y}_{t',i'} - \beta_0 - \boldsymbol{x}_{t',i'}^\top \boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{p} \omega_j \, \beta_j^2,$$

$$\hat{\boldsymbol{\beta}}_{t,i}^{\text{enet}} = \underset{\beta_0, \boldsymbol{\beta}}{\arg\min} \sum_{(t',i') \in \mathcal{W}_{t,i}} (\tilde{y}_{t',i'} - \beta_0 - \boldsymbol{x}_{t',i'}^\top \boldsymbol{\beta})^2 + \lambda\alpha \sum_{j=1}^{p} \omega_j |\beta_j| + \frac{\lambda(1-\alpha)}{2} \sum_{j=1}^{p} \omega_j \, \beta_j^2,$$

  with $\lambda > 0$ the coefficient on the penalty, $\alpha \in [0, 1]$ the elastic-net mixing weight, and $\omega_j \geq 0$ the penalty weight on predictor j (intercept β_0 unpenalized, $\omega_0 = 0$). For ridge, λ is the value in Table 4. For lasso and elastic net that table prints the scikit-learn parameter, and the coefficient in the sum is $2n$ times the printed value, with n the number of rows in the sum.
  - **Block penalty** (block-penalized ridge, lasso and elastic net): the HAR benchmark is penalized 100 times more lightly than the additional predictors,

$$\omega_j = \begin{cases} \underline{\omega} = 0.01 & j \in \text{HAR benchmark}, \\ 1 & j \text{ an additional predictor}, \\ 0 & j = 0 \text{ (intercept)}. \end{cases}$$

  - HAR benchmark alone: $\omega_j = 1$ for every predictor, a single penalty.
  - Lasso: $\alpha = 1$. Elastic net: $\alpha = 0.5$.
  - Ridge: rank-one add and drop on $(\boldsymbol{X}_{t,i}^\top \boldsymbol{X}_{t,i} + \lambda \, \text{diag}(\boldsymbol{\omega}))^{-1}$, where $\boldsymbol{X}_{t,i}$ stacks the rows $(1, \boldsymbol{x}_{t',i'}^\top)$, $(t', i') \in \mathcal{W}_{t,i}$, and $\boldsymbol{\omega} = (\omega_0, \ldots, \omega_p)^\top$.
  - Lasso, elastic net: exact online homotopy (Garrigues and El Ghaoui, 2008), one observation in, one out.
  - At every fit, only the screened columns $\mathcal{J}_{t,i}$ enter: predictors constant within the window or exact copies of another column are dropped; in a model estimated on the closing interval only, this always removes hour and the overnight indicator (Section 4).

- **Tree ensembles:** gradient boosting (LightGBM, XGBoost) and random forest.
  - **Gradient boosting** (Friedman, 2001), in LightGBM (Ke et al., 2017) and XGBoost (Chen and Guestrin, 2016): squared-error loss $\frac{1}{2}(\tilde{y} - F)^2$, rounds $b = 1, \ldots, B$, learning

rate η. The tree f_b has J leaves and leaf weights $\boldsymbol{w}$; leaf j holds the observations $\mathcal{R}_j$, and $n_j = |\mathcal{R}_j|$. At $\lambda_1^{\text{tree}} = 0$ the fitted function and the leaf weight are

$$F_b = F_{b-1} + \eta f_b,$$
$$w_j^* = -\frac{\sum_{(t',i')\in\mathcal{R}_j}\big(F_{b-1}(\boldsymbol{x}_{t',i'}) - \tilde{y}_{t',i'}\big)}{n_j + \lambda_2^{\text{tree}}}.$$

The tree minimizes

$$\sum_{(t',i')\in\mathcal{W}_{t,i}} \Big[\big(F_{b-1}(\boldsymbol{x}_{t',i'}) - \tilde{y}_{t',i'}\big) f_b(\boldsymbol{x}_{t',i'}) + \tfrac{1}{2} f_b(\boldsymbol{x}_{t',i'})^2\Big] + \gamma^{\text{tree}} J + \lambda_1^{\text{tree}}\|\boldsymbol{w}\|_1 + \tfrac{1}{2}\lambda_2^{\text{tree}}\|\boldsymbol{w}\|_2^2.$$

– γ^{tree}: penalty per leaf; $\lambda_1^{\text{tree}}, \lambda_2^{\text{tree}}$: L1 and L2 penalties on $\boldsymbol{w}$.
– Leaf minimum $n_{\min}$: $n_j \geq n_{\min}$. That count is LightGBM's minimum leaf size and, under this loss, XGBoost's minimum child weight.
– **Random forest** (Breiman, 2001): average of B trees f_b, each grown on a bootstrap sample,

$$\hat{y}_{t,i} = \frac{1}{B}\sum_{b=1}^{B} f_b(\boldsymbol{x}_{t,i});$$

a random subset of the predictors is tried at each split.

5.3 Forecast evaluation

- **Forecast and realization.** The variance forecast is the back-transform of Section 5.1, with no bias correction. The realization is the actual realized variance of the closing interval, $\text{RV}_t \equiv \text{RV}_{t,c}$; it is not winsorized (the winsorized $\tilde{y}_{t,c}$ enters estimation only).
- **Evaluation set.** Every forecast comes from a model re-estimated on its trailing window, so every forecast is out of sample. $\mathcal{T}_{\text{oos}}$ is the set of forecast origins, trading days, after the initial estimation window and after the pre-sample on which hyperparameters are selected, on which every forecast compared exists; early-close days are excluded, and $T_{\text{oos}} = |\mathcal{T}_{\text{oos}}|$ (its dates and T_{oos}: Experimental setup, Section 6.1).
- **Loss.** The daily QLIKE loss (Patton, 2011), robust to noise in the realized-variance proxy. After the diurnal adjustment cancels ($s_{t,c}^2$, with the unwinsorized y_t),

$$L_t = \frac{y_t^2}{\hat{y}_t^2} - \log\frac{y_t^2}{\hat{y}_t^2} - 1 = \text{QLIKE}(\text{RV}_t,\, \hat{\sigma}_t^2),$$

and its average over $\mathcal{T}_{\text{oos}}$:

$$\overline{\text{QLIKE}} = \bar{L} = \frac{1}{T_{\text{oos}}}\sum_{t\in\mathcal{T}_{\text{oos}}} L_t.$$

- **Benchmark forecast.** The daily HAR regression (Corsi, 2009): OLS on the HAR benchmark, the HAR components and the calendar dummies (Section 5.2), estimated on the closing interval only, one observation per trading day, and re-estimated every trading day.
- **Relative loss.** For forecast A against forecast B (the benchmark forecast unless stated otherwise), the percentage change in average QLIKE is

$$\Delta\% = 100\Big(\frac{\bar{L}^A}{\bar{L}^B} - 1\Big).$$

Its significance is the Diebold–Mariano statistic of Appendix H.

- **Squared-error and absolute-error loss,** in variance units:

$$\mathrm{MSE} = \frac{1}{T_{\mathrm{oos}}} \sum_{t \in \mathcal{T}_{\mathrm{oos}}} \left(\mathrm{RV}_t - \hat{\sigma}_t^2\right)^2, \qquad \mathrm{MAE} = \frac{1}{T_{\mathrm{oos}}} \sum_{t \in \mathcal{T}_{\mathrm{oos}}} \left|\mathrm{RV}_t - \hat{\sigma}_t^2\right|.$$

- **Out-of-sample** R^2 (Campbell and Thompson, 2008), with the historical mean of the realized variance, known at the forecast origin, as the reference forecast:

$$R^2_{\mathrm{OOS}} = 1 - \frac{\sum_{t \in \mathcal{T}_{\mathrm{oos}}} \left(\mathrm{RV}_t - \hat{\sigma}_t^2\right)^2}{\sum_{t \in \mathcal{T}_{\mathrm{oos}}} \left(\mathrm{RV}_t - \overline{\mathrm{RV}}_{t-1}\right)^2}, \qquad \overline{\mathrm{RV}}_{t-1} = \frac{1}{|\{t' < t\}|} \sum_{t' < t} \mathrm{RV}_{t'},$$

  the expanding mean over every earlier trading day (start in Section 6.1).
- **Calibration:** the ratio of average realized to average forecast variance, $\overline{\mathrm{RV}}/\overline{\hat{\sigma}^2}$, and the frequency of under-prediction, $\frac{1}{T_{\mathrm{oos}}} \sum_{t \in \mathcal{T}_{\mathrm{oos}}} \mathbf{1}\{\mathrm{RV}_t > \hat{\sigma}_t^2\}$.

6 Experiments and results

6.1 Experimental setup

- **Evaluation span** $\mathcal{T}_{\mathrm{oos}}$ (Section 5.3): the rolling forecasts from 3 January 2020 to 30 April 2024, $T_{\mathrm{oos}} = |\mathcal{T}_{\mathrm{oos}}| = 1{,}082$ trading days, every trading day on which all forecasts exist (6 early-close days dropped). It starts with the intraday linear forecasts, after the pre-sample on which the linear models' penalties are selected (6 January 2016 to 2 January 2020).
- **Training scheme:** intraday, all $M = 48$ intervals of each day; $W = 24{,}000$ intervals for the trees and 96,000 for the linear models. On the full predictor set, the pooled LightGBM and XGBoost are the earlier draft's specification and the pooled random forest is preliminary (Table 1).

  E15. Decision: one hyperparameter rule for every class, frozen after a single pre-sample pick.
  Problem: the sample and the window are still open, and the printed QLIKE cells for the linear light re-tune, the HAR-benchmark λ path, the full-set LightGBM draft, the full-set XGBoost draft, and the preliminary random forest were not produced by one frozen configuration.
  The rule for the comparison is one configuration per model, chosen once on the pre-sample and then held fixed for the whole evaluation span. Two inputs to that rule are still unset, and the only write-up of the comparison is in the author notes. The sample is either every one of the 48 intervals or the 13 regular-hours bars whose end stamps run from 10:00 to 16:00. The window is either $W = 24{,}000$ intervals or $W = 96{,}000$ intervals. On top of that open choice, some cells already printed in Table 1 were not produced by one frozen configuration. The linear penalties still move under the light re-tune, the HAR-benchmark penalties move on the dates left open in note E23, and the full-set LightGBM row, the full-set XGBoost row, and the random-forest row are an earlier draft or a preliminary fit. Those QLIKE numbers are not the losses of one shared configuration, and the table has not been refit.
- **Benchmark forecast:** the daily HAR regression, $W = 2{,}000$ trading days; the models trained the same way, on the last 30 minutes of each day, are in Section 7.4.
- **Re-estimated every period** (Appendix B.2); hyperparameters are fixed or selected causally, on a pre-sample or on validation samples that precede the forecast origin.
- **Loss and tests:** QLIKE against the actual realized variance (Section 5.3); Diebold–Mariano tests with the Harvey–Leybourne–Newbold correction ($q_{\mathrm{NW}} = 6$ on $\mathcal{T}_{\mathrm{oos}}$), with a Newey–West long-run variance.[1]

Full implementation details are in Appendix B.

[1] These results have a more rigorous statistical analysis in Appendix H.

6.2 Out-of-sample accuracy

- **Overview:** Table 1 reports the out-of-sample MSE, average QLIKE and R^2_{OOS} of every intraday model, on the HAR benchmark and on all features, over the 1,082 trading days of the evaluation span $\mathcal{T}_{\text{oos}}$. The same statistics for the models trained on the last 30 minutes, whose OLS on the HAR benchmark is the benchmark forecast, are in Table 2 (Section 7.4).
- R^2_{OOS} follows Campbell and Thompson (2008): the benchmark is the historical mean of the closing-half-hour realized variance from 5 January 1998 through the day before the forecast, so it uses no out-of-sample information.

6.3 The full predictor set versus the HAR benchmark

- **Paired comparison** within each model class: the same intraday model m ($\mathcal{I} = \{1, \ldots, M\}$) on the two predictor sets, $d_t = L_t^{\text{Full}} - L_t^{\text{HAR}}$, with $L_t^{\mathcal{X}}$ the loss of the forecast $f_{m,\mathcal{X},\{1,\ldots,M\}}$. $\Delta\%$ is the relative loss of Section 5.3; DM is on d_t over $\mathcal{T}_{\text{oos}}$, with $q_{\text{NW}} = 6$.
- **The full predictor set lowers average QLIKE in every model class** ($\Delta\% < 0$ in all 6), 4 of them significantly at the 5% level (Table 1):
  - trees, the largest reductions: $\Delta\% = -13.0$, -12.0 and -11.9 for LightGBM, XGBoost and the random forest (DM $= -4.39$, -4.29, -4.75), each significant;
  - linear models: elastic net $\Delta\% = -5.6$ (DM $= -2.10$, significant), ridge -5.0 (DM $= -1.77$, $p = 0.077$), lasso -3.0 (DM $= -1.13$, $p = 0.26$).
- **Squared-error loss does not confirm the reductions:** the full predictor set raises the MSE in 3 of the 6 classes and lowers it in the others, none significantly (DM on the squared error).
  - Each pooled forecast's five largest squared errors on the full predictor set fall in March 2020 and carry 72–83% of its $\sum_{t\in\mathcal{T}_{\text{oos}}}(\text{RV}_t - \hat{\sigma}_t^2)^2$, but only 1–4% of its total QLIKE loss.

6.4 Model classes

- **The linear models are the most accurate intraday forecasts on each predictor set** (Table 1):
  - full predictor set: average QLIKE 0.1078–0.1101 for the linear models (elastic net 0.1078, ridge 0.1086, lasso 0.1101, the three most accurate of the 12 pooled forecasts) and 0.1126–0.1146 for the trees;
  - HAR benchmark: 0.1135–0.1143 for the linear models and 0.1279–0.1301 for the trees, the least accurate pooled forecasts.
- **Every intraday forecast has a higher average QLIKE than the benchmark forecast** ($\overline{\text{QLIKE}} = 0.1033$; $\Delta\%$ and DM against it):
  - full predictor set: linear models +4.4 to +6.7%, none significant at the 5% level; trees +9.0 to +11.0%, each significant at the 5% level;
  - HAR benchmark: +10.0 to +26.0%, each significant at the 1% level;
  - the full predictor set narrows the difference in every model class;
  - the benchmark forecast is trained on the last 30 minutes of each day; Section 7.4 trains every model class that way.

Table 1: Forecast accuracy of the intraday models

	Intraday		
Model and predictor set	MSE (10^{-10})	QLIKE	R^2_{OOS} (%)
Panel A: Linear models			
OLS			
HAR benchmark[a]	4.82	0.1291	74.7
Ridge			
HAR benchmark	4.74	0.1143	75.1
All features	4.79	0.1086	74.9
Lasso			
HAR benchmark	4.91	0.1135	74.2
All features	4.70	0.1101	75.4
Elastic net			
HAR benchmark	4.94	0.1142	74.1
All features	4.72	0.1078	75.3
Panel B: Tree ensembles			
LightGBM			
HAR benchmark	3.98	0.1296	79.1
All features[a]	4.13	0.1128	78.3
XGBoost			
HAR benchmark	3.85	0.1279	79.8
All features[a]	4.01	0.1126	79.0
Random forest			
HAR benchmark	3.92	0.1301	79.4
All features[†]	3.75	0.1146	80.3

Notes: Rolling forecasts from 3 January 2020 to 30 April 2024, every forecast made with a model estimated on its trailing window only, so every forecast is out of sample; each is scored against the actual realized variance of the closing half hour, RV_t, on the $T_{\text{oos}} = 1{,}082$ trading days on which every forecast of both training schemes exists, early closes dropped (Section 5.3). The span starts with the intraday linear forecasts, on the first trading day after the pre-sample on which the penalties were selected (6 January 2016 to 2 January 2020). The MSE entries are in units of 10^{-10}: the printed number times 10^{-10} is the mean squared error $T_{\text{oos}}^{-1}\sum_t(\text{RV}_t - \hat{\sigma}_t^2)^2$. QLIKE is the mean QLIKE loss. $R^2_{\text{OOS}} = 1 - \sum_t(\text{RV}_t - \hat{\sigma}_t^2)^2 / \sum_t(\text{RV}_t - \overline{\text{RV}}_{t-1})^2$, in percent, where the benchmark $\overline{\text{RV}}_{t-1}$ is the historical mean of the closing-half-hour realized variance over all trading days from 5 January 1998 to day $t-1$ (Campbell and Thompson, 2008). All features is the full predictor set of Section 4. The same days for the models trained on the last 30 minutes are in Table 2 (Section 7.4).
[a] The earlier draft's intraday specification.
[†] Preliminary: the closing-interval forecasts of the intraday random forest on all features, from an exercise still in progress.

6.5 Variable importance

- **Models:** the closing-interval models of record on the full predictor set ($p = 628$), trained on the last 30 minutes of each trading day (Section 7.4), not on the intraday scheme.
  - Lasso and ridge with the penalty strength λ selected on a pre-sample: $\lambda = 10^{-2}$ and 10^{3} (scikit-learn parametrization). Penalty weight ω_j, the factor on predictor j's penalty term: $\underline{\omega} = 0.01$ on the HAR benchmark block, 1 on the exogenous predictors. Re-estimated every trading day; the scheduled re-selection of λ never replaced the pre-sample choice.
  - Not the earlier study's single-penalty specification (one λ for every predictor, re-selected every 250 trading days): the paper does not use it.
  - Tree ensembles at fixed hyperparameters, the best entries of the development-sample search (LightGBM and XGBoost: the top Optuna trial; random forest: trial 255); constant and duplicate predictors dropped in each estimation window.
- **Notation.**
  - Re-estimation $k = 1, \ldots, K$, $K = 147$, one every 10 trading days. Re-estimation k uses the window $\mathcal{W}_{t_k}$, the 2,000 trading days before its first forecast day t_k, and forecasts its out-of-sample block $\mathcal{O}_k$, the trading days from t_k to the next re-estimation (≤ 10). The blocks make up this section's forecast days $\mathcal{O} = \bigcup_k \mathcal{O}_k$: $|\mathcal{O}| = 1{,}469$ trading days, 25 June 2018 to 30 April 2024.

    E19. Decision: one cadence, and label the sample on every rank.
    Problem: the model bullet refits every trading day, the notation refits every 10 trading days, and a rank on 25 June 2018–30 April 2024 comes from the closing-interval fits rather than from Table 1. The model bullet says lasso and ridge are re-estimated every trading day, and that λ stays at the pre-sample values 10^{-2} and 10^{3}. The notation bullet then counts $K = 147$ re-estimations, spaced 10 trading days apart, each one fit on the 2,000 days before its first forecast day t_k and then held for the block of days up to the next re-estimation. A fit every day and a fit every 10 days produce different coefficients on the days in between. The days those fits are asked to forecast run from 25 June 2018 to 30 April 2024, $|\mathcal{O}| = 1{,}469$ days in total. Table 1 is another sample: the intraday scheme on $T_{\text{oos}} = 1{,}082$ days, from 3 January 2020 to 30 April 2024. A rank in this section is a summary of the closing-interval fits on 25 June 2018–30 April 2024. It is not a row of that table.

  - Predictors $\boldsymbol{x}_t \in \mathbb{R}^p$; forecast $\hat{y}_t = \hat{f}_k(\boldsymbol{x}_t)$, $t \in \mathcal{O}_k$, on the scale of the target y_t, the diurnally adjusted realized volatility of the closing half hour.
  - Predictor group $A \subseteq \{1, \ldots, p\}$: all predictors built from one series (its averages at every horizon, its missing-data and activity indicators). 49 groups: 47 exogenous series in 8 categories (Section 4), the HAR components, the calendar dummies.
  - Linear models: $\hat{\boldsymbol{\beta}}_k \in \mathbb{R}^p$, the coefficient estimates behind the forecast for t_k; $\hat{\boldsymbol{\beta}}_{k,A}$, its subvector on group A.
  - Trees: SHAP values $\phi_{t,j}$ (Lundberg and Lee, 2017) and SHAP interaction values $\phi_{t,jl}$ (Lundberg et al., 2020); they add up to the forecast,

$$\hat{y}_t = \phi_0 + \sum_{j=1}^{p} \phi_{t,j}, \qquad \phi_{t,j} = \sum_{l=1}^{p} \phi_{t,jl}, \qquad \phi_{t,A} = \sum_{j \in A} \phi_{t,j}.$$

    For $j \neq l$, $\phi_{t,jl} = \frac{1}{2}(\phi_{t,l} \mid j \text{ present} - \phi_{t,l} \mid j \text{ absent})$, and $\phi_{t,jj} = \phi_{t,j} - \sum_{l \neq j} \phi_{t,jl}$.
  - Path-dependent TreeSHAP (Lundberg et al., 2020): the training observations per node (node covers) stand in for the background sample.
  - Number triples below: LightGBM, XGBoost, random forest.

- **Lasso, aggregate:** the selection count of a group and of a predictor, the number of the K re-estimations with a nonzero coefficient,

$$K_A = \sum_{k=1}^{K} \mathbf{1}\{\exists j \in A : \ \hat{\beta}_{k,j} \neq 0\}, \qquad K_j = \sum_{k=1}^{K} \mathbf{1}\{\hat{\beta}_{k,j} \neq 0\};$$

  K_A/K is the selection frequency, the share of estimation windows with a nonzero coefficient.
  - Per re-estimation: 30 nonzero coefficients (median; range 27–35), of which 14 on exogenous predictors.
  - Predictors: 60 of 628 selected at least once, 29 in at least half of the re-estimations.
  - Exogenous series: 27 of 47 selected at least once; 11 in at least half of the re-estimations; 3 in every re-estimation (r, $\mathrm{RAV}^{\mathrm{VW}}$, AC).
  - HAR benchmark block: selected in every re-estimation ($\omega_j = 0.01$); 88 % of $\sum_j |\hat{\beta}_{k,j}|$ (median re-estimation).
  - Random-probe benchmark: the continuous probe, an i.i.d. uniform noise predictor (defined under Known issues), is selected in 2.0 % of its fits; 54 predictors, in 23 exogenous series, are selected more often.
- **Lasso, time-varying:** the magnitude and the selection indicator of each group at each re-estimation,

$$\|\hat{\boldsymbol{\beta}}_{k,A}\|_1 = \sum_{j \in A} |\hat{\beta}_{k,j}|, \qquad \mathbf{1}\{\|\hat{\boldsymbol{\beta}}_{k,A}\|_1 > 0\}$$

  Rank stability across re-estimations: Kendall's coefficient of concordance W of the rankings by $\|\hat{\boldsymbol{\beta}}_{k,A}\|_1$ at each re-estimation ($W = 1$: the same ranking at every re-estimation; formula below), $W = 0.74$.
- **Ridge, aggregate:** the group's share of the absolute coefficient mass,

$$S_A = \frac{\sum_k \sum_{j \in A} |\hat{\beta}_{k,j}|}{\sum_k \sum_{j=1}^{p} |\hat{\beta}_{k,j}|}.$$

  - Scale check: only the averages of series values are robust-scaled.
    - Averages of series values (282 predictors): standard deviation over the estimation window 0.84 (median; 5th–95th percentiles: 0.03–1.76), interquartile range 0.94.
    - Averaged missing-data and activity indicators (324): standard deviation 0.022; calendar dummies 0.39; HAR components 1.07.
    - So the raw $|\hat{\beta}_{k,j}|$ are in different units. Check with the group's share of the absolute standardized coefficients,

$$S_A^{\mathrm{sd}} = \frac{\sum_k \sum_{j \in A} |\hat{\beta}_{k,j}| \, \mathrm{sd}_{k,j}}{\sum_k \sum_{j=1}^{p} |\hat{\beta}_{k,j}| \, \mathrm{sd}_{k,j}},$$

      with $\mathrm{sd}_{k,j}$ the standard deviation of predictor j over $\mathcal{W}_{t_k}$; Spearman rank correlation of S_A^{sd} with S_A across groups 0.85.
    - Leading groups on that scale: HAR components 30.6 %, calendar dummies 9.3 %, r 4.5 %.
  - Leading groups: calendar dummies 31.0 %, HAR components 22.8 %, r 3.6 %; HAR benchmark block 54 % (median re-estimation).

  - Ridge sets no coefficient to zero: all 49 groups have $K_A = K$, so the selection frequency does not discriminate.
  - Random-probe benchmark: 107 predictors, in 38 exogenous series, have $|\hat{\beta}_{k,j}|$ above the continuous probe's (robust-scaled) in most probe fits.
- **Ridge, time-varying:** the share at re-estimation k, $S_{k,A} = \|\hat{\boldsymbol{\beta}}_{k,A}\|_1 / \sum_{A'} \|\hat{\boldsymbol{\beta}}_{k,A'}\|_1$, A' over all groups (mean rank by $\|\hat{\boldsymbol{\beta}}_{k,A}\|_1$); Kendall's $W = 0.85$.
- **Trees, ranking at each re-estimation:** the group's mean absolute SHAP value on the out-of-sample block of re-estimation k, then its rank,

$$I_{k,A} = \frac{1}{|\mathcal{O}_k|} \sum_{t \in \mathcal{O}_k} |\phi_{t,A}|, \qquad r_{k,A} = 1 + \#\{A' : I_{k,A'} > I_{k,A}\} \quad \text{(ties averaged)}.$$

- **Trees, global ranking:** aggregate the K rankings over time,

$$\bar{r}_A = \frac{1}{K} \sum_{k=1}^{K} r_{k,A}, \qquad \mathrm{BC}_A = \sum_{k=1}^{K} \max(0,\ 11 - r_{k,A}).$$

  - BC_A: truncated Borda count (a full Borda count is affine in $\bar{r}_A$).
  - Kendall's coefficient of concordance (Kendall and Babington Smith, 1939) over the K rankings of the n groups, with rank sums $R_A = \sum_k r_{k,A}$ and their mean $\bar{R}$:

$$W = \frac{12 \sum_A (R_A - \bar{R})^2}{K^2(n^3 - n) - K \sum_k \sum_{g \in \mathcal{E}_k} (|g|^3 - |g|)},$$

    $\mathcal{E}_k$ the sets of tied (equally ranked) groups at re-estimation k; $W = 1$: the same ranking at every re-estimation.
  - LightGBM: HAR components ($\bar{r}_A = 1.0$), r (2.9), RAV (4.8); Kendall's $W = 0.57$.
  - XGBoost: HAR components (1.0), r (3.0), RAV (3.9); Kendall's $W = 0.60$.
  - Random forest: HAR components (1.0), RAV (2.6), r (4.0); Kendall's $W = 0.62$.
  - Mean rank vs truncated Borda count, Spearman rank correlation over groups: 0.90, 0.87, 0.92.
  - Aggregate of the ranks vs rank of the aggregate, the mean absolute SHAP value over $\mathcal{O}$, $\bar{I}_A = |\mathcal{O}|^{-1} \sum_{t \in \mathcal{O}} |\phi_{t,A}|$: 0.96, 0.97, 0.98.
  - Predictor level ($n = 628$): Kendall's $W = 0.91$, 0.90, 0.93.
  - Agreement of $\bar{r}_A$ across models (Spearman rank correlation): trees 0.83–0.96; tree vs linear 0.17–0.55; lasso vs ridge 0.47.
- **Interaction effects, global:** the mean absolute SHAP interaction value of two groups over $\mathcal{O}$,

$$\phi_{t,AB} = \sum_{j \in A} \sum_{l \in B} (\phi_{t,jl} + \phi_{t,lj}), \qquad \bar{I}_{AB} = \frac{1}{|\mathcal{O}|} \sum_{t \in \mathcal{O}} |\phi_{t,AB}|, \quad A \neq B;$$

  for two predictors, $|\phi_{t,jl} + \phi_{t,lj}|$, $j \neq l$.
  - Top pair: LightGBM HAR components × RAV (12.6); XGBoost HAR components × RAV (15.2); random forest HAR components × RAV (23.8); $\times 10^{-3}$, in units of y_t.
  - Scale: largest $\bar{I}_A$ 195, 192, 224 ($\times 10^{-3}$); all series-pair $\bar{I}_{AB}$ summed, as % of $\sum_A \bar{I}_A$: 90, 74, 72.
  - Within a group ($\sum_{j \neq l \in A} \phi_{t,jl}$): largest HAR components (18.7), HAR components (19.1), HAR components (34.3); $\times 10^{-3}$.

  - Noise floor: the continuous probe's largest $\bar{I}$ with any group, median over probe fits: 0.49, 0.41, 0.50 ($\times 10^{-3}$); series pairs above it: 116, 115, 52 of 1,176.

- **Interaction effects, cross-check:** the H-statistic of Friedman and Popescu (2008) on each model's top 10 series pairs, from the partial dependence functions PD_S, $S \in \{A, B, AB\}$, $AB \equiv A \cup B$:

$$H^2_{AB} = \frac{\sum_t \left[\mathrm{PD}_{AB}(\boldsymbol{x}_t) - \mathrm{PD}_A(\boldsymbol{x}_t) - \mathrm{PD}_B(\boldsymbol{x}_t)\right]^2}{\sum_t \mathrm{PD}_{AB}(\boldsymbol{x}_t)^2}, \qquad \mathrm{PD}_S(\boldsymbol{x}) = \frac{1}{|\mathcal{H}_k|} \sum_{t' \in \mathcal{H}_k} \hat{f}_k(\boldsymbol{x}_S, \boldsymbol{x}_{t',-S}),$$

  $t \in \mathcal{H}_k$; each PD_S centered over $\mathcal{H}_k$.
  - Points: $\mathcal{H}_k$, the 100 trading days from t_k on, both evaluation points and background sample, at 15 re-estimations (every tenth); sums pooled over re-estimations.
  - The denominator contains the HAR components' own partial dependence whenever they are in the pair. A second normalization, by the smaller single-group partial dependence:

$$H^2_{\min,AB} = \frac{\sum_t \left[\mathrm{PD}_{AB}(\boldsymbol{x}_t) - \mathrm{PD}_A(\boldsymbol{x}_t) - \mathrm{PD}_B(\boldsymbol{x}_t)\right]^2}{\min\{\sum_t \mathrm{PD}_A(\boldsymbol{x}_t)^2,\ \sum_t \mathrm{PD}_B(\boldsymbol{x}_t)^2\}}.$$

  - Top pair, H^2 (%): 0.33, 0.37, 0.85; over the top 10: 0.02–0.33, 0.02–0.68, 0.02–0.85.
  - Top pair, $H^2_{\min}$: 0.19, 0.39, 1.50; over the top 10: 0.05–0.54, 0.08–0.80, 0.17–5.46.
  - Top predictor pair, H^2 (%) and $H^2_{\min}$: LightGBM 0.36, 0.02; XGBoost 0.39, 0.02; random forest 1.16, 0.06.

- **Interaction effects, time-varying:** $\phi_{t,AB}$ on the out-of-sample block of each re-estimation, ranked at each re-estimation and aggregated as the SHAP values are,

$$I_{k,AB} = \frac{1}{|\mathcal{O}_k|} \sum_{t \in \mathcal{O}_k} |\phi_{t,AB}| \ \rightarrow\ r_{k,AB} \ \rightarrow\ \bar{r}_{AB},\ W.$$

  - Kendall's W over each model's top 50 pairs: 0.45, 0.47, 0.57.
  - Mean rank vs $\bar{I}_{AB}$ (Spearman rank correlation, all pairs): 0.99, 0.99, 1.00.

- **Known issues, reconciled:**
  - Mean decrease in impurity (MDI) favors predictors with many distinct values (Strobl et al., 2007): MDI is not used; SHAP values and coefficients only.
  - Random probes, at every third re-estimation (49 fits): three i.i.d. noise predictors, continuous (U(0, 1)), discrete uniform on 20 levels, and binary (Bernoulli, probability one half).
    * Lasso: the probes are selected in 1, 0, 0 of 49 fits (robust-scaled, $\omega_j = 1$).
    * Ridge: 107 predictors have $|\hat{\beta}_{k,j}|$ above the continuous probe's in most probe fits.
    * Trees: the continuous probe's median rank among 631 predictors by mean $|\phi_{t,j}|$: 83, 110, 97; exogenous series with a predictor above it in most probe fits: 27, 35, 29.
  - Dilution across correlated predictors: $\phi_{t,A}$ and $\|\hat{\boldsymbol{\beta}}_{k,A}\|_1$ sum over the group's predictors, so credit split among a series' averages at different horizons stays inside the group.
  - The earlier study's single-penalty lasso (λ re-selected): median 23 nonzero coefficients per re-estimation; 18 exogenous series selected in at least half of the re-estimations; Spearman rank correlation of its K_A with that of the lasso of record 0.70.
  - Reproduction: re-running the estimation reproduces the archived LightGBM models exactly; XGBoost and the random forest differ across library versions (maximum absolute forecast difference 0.17 and 0.11, in units of y_t). SHAP values: from the archived

models; interaction values: from the reproduced ones (Spearman rank correlation of $\bar{r}_A$, reproduced vs archived: 1.000, 0.993, 0.996).

- **Compute** (one laptop, 11 processes):
  - Reproducing the tree models and computing the SHAP interaction values: 133 minutes. Per re-estimation: random forest estimation 212 s, its SHAP and interaction values 182 s; LightGBM 0.9 s, XGBoost 0.4 s.
  - Linear coefficient paths and probes: 7 minutes; H^2: 7 minutes.

7 Economic value of the forecasts

7.1 The straddle strategy

- **Position:** SPXW options that expire the same day (0DTE). With S_t the underlying print on the option chain at the 15:30 ET forecast origin, the call and the put at the nearest strikes on either side,

$$K_C = \min\{K \geq S_t\}, \qquad K_P = \max\{K \leq S_t\},$$

over strikes with live quotes. If $K_C = K_P = S_t$ the package is an at-the-money straddle; otherwise it is a strangle on the two strikes adjacent to S_t. We call both the straddle: a position that is approximately delta-neutral at entry, entered at 15:30 ET and held, not hedged, to expiration.

E18. Decision: write the day filter into the text, and give the implied-volatility line the recovery used on the censored days.
Problem: the day-drop rule (the wider strike more than 10 index points from S_t, or fewer than 10 quoted contracts) is still only a comment under this item, and the implied-volatility line and Appendix C make a censored leg missing, while the scored days recover it.
S_t is now the 15:30 underlying print on the option chain and S_t^{close} the official S&P 500 close. The notebook checks that the Yahoo closes match the settlement column on every scored day. The chain field is copied as printed, and the code never labels it cash or futures, so the text calls it neither. A day is dropped when the wider strike is more than 10 index points from S_t, or when fewer than 10 contracts are quoted. That rule is in Appendix C and is still only a comment under this item. Early-close days are a separate drop, already in the T_{opt} line of the Data section. The vendor field is an hourly standard deviation from a bisection on [0.0005, 0.025]. Dividing by $\sqrt{2}$ is the half-hour conversion, and that is what was scored. On the 5 scored days where a leg is a failed-solve node, the hourly volatility is recovered by inverting the Black–76 price of the straddle on the 15:30 underlying, then divided by $\sqrt{2}$. Appendix C says a censored leg makes the implied variance missing. That is the drop path, not the scored path.

- **Entry prices:** at the mid-quote, P_t, and at the quoted bid and ask of the package, P_t^b and P_t^a, from the legs' bid quotes b and ask quotes a at 15:30 ET:

$$P_t = \tfrac{1}{2}(b_t^C + a_t^C) + \tfrac{1}{2}(b_t^P + a_t^P), \qquad P_t^b = b_t^C + b_t^P, \qquad P_t^a = a_t^C + a_t^P.$$

- **Settlement** in cash at the official S&P 500 close S_t^{close}, with no delta hedging; R_t is the straddle's simple return on the premium paid at the mid-quote:

$$X_t = (S_t^{\text{close}} - K_C)^+ + (K_P - S_t^{\text{close}})^+, \qquad R_t = \frac{X_t}{P_t} - 1 \;\geq\; -1.$$

- **Implied variance:** IV_t^C and IV_t^P are the legs' implied volatilities at 15:30 ET, quoted per hour; the implied volatility of the closing half hour is

$$\text{IV}_t = \frac{\text{IV}_t^C + \text{IV}_t^P}{2\sqrt{2}},$$

and IV_t^2 is the implied variance to expiration.

- **Forecast variance risk premium:** the variance risk premium of the closing half hour is $\mathrm{IV}_t^2 - \mathbb{E}_t[\mathrm{RV}_t]$ (Carr and Wu, 2009; Bollerslev et al., 2009), and its forecast is $\mathrm{IV}_t^2 - \hat{\sigma}_t^2$, with $\hat{\sigma}_t^2$ the variance forecast of Section 5.1. Every strategy below conditions on the forecast only through the gap $\hat{\sigma}_t^2 - \mathrm{IV}_t^2$: it buys when the forecast exceeds implied variance.
- **Strategies:** position q_t in straddles ($q_t > 0$ long, $q_t < 0$ short) and daily return $r_t = q_t R_t$. The forecast enters only through q_t, so the comparison across strategies measures the economic value of the forecast (Fleming et al., 2001):
  - the unconditional short-straddle benchmark, a static short-variance position: $q_t \equiv -1$;
  - sign strategy: $q_t = \mathrm{sign}(\hat{\sigma}_t^2 - \mathrm{IV}_t^2)$, with $q_t = -1$ on ties;
  - robust clip: $q_t = \mathrm{clip}\big((\hat{\sigma}_t^2 - \mathrm{IV}_t^2)/\mathrm{MAD}_t, -1, 1\big)$, with the scale $\mathrm{MAD}_t = \mathrm{median}_{t'<t}|\hat{\sigma}_{t'}^2 - \mathrm{IV}_{t'}^2|$, the median absolute gap over the earlier days of $\mathcal{T}_{\mathrm{opt}}$;
  - full long, scaled short: $q_t = 1$ if $\hat{\sigma}_t^2 > \mathrm{IV}_t^2$, else $q_t = \max\big((\hat{\sigma}_t^2 - \mathrm{IV}_t^2)/\mathrm{MAD}_t, -1\big)$;
  - rank of the absolute gap, over the $t-1$ earlier days $t' < t$ of $\mathcal{T}_{\mathrm{opt}}$:

$$q_t = \mathrm{sign}(\hat{\sigma}_t^2 - \mathrm{IV}_t^2) \cdot \frac{1}{t-1} \sum_{t'<t} \mathbf{1}\{|\hat{\sigma}_{t'}^2 - \mathrm{IV}_{t'}^2| \le |\hat{\sigma}_t^2 - \mathrm{IV}_t^2|\}.$$

- **Sizing sample:** q_t uses days $t' < t$ only. The first 252 days of $\mathcal{T}_{\mathrm{opt}}$ build the scale MAD_t, so every sizing rule, the sign strategy included, is evaluated on the remaining 614 trading days, $t > 252$.
- **Execution:** at the mid-quote, $r_t = q_t R_t$. At quoted bid/ask prices a purchase pays the ask and a sale receives the bid; the spread is paid at entry only, since settlement is in cash:

$$r_t^{\mathrm{ba}} = \begin{cases} |q_t|\,(X_t/P_t^a - 1), & q_t > 0, \\ |q_t|\,(1 - X_t/P_t^b), & q_t < 0. \end{cases}$$

- **Transaction costs:** trading at the quoted bid or ask instead of the mid-quote costs a median 2.9% of the premium. No other transaction costs are deducted.

7.2 Performance measures

- **Annualized Sharpe ratio,** from daily returns, at mid-quotes and at quoted bid/ask prices, over the 614 days of $\mathcal{T}_{\mathrm{opt}}$ with $t > 252$:

$$\mathrm{SR} = \sqrt{252}\,\frac{\bar{r}}{\mathrm{sd}(r)}, \qquad \mathrm{sd}(r)^2 = \frac{1}{614-1} \sum_{\substack{t \in \mathcal{T}_{\mathrm{opt}} \\ t>252}} (r_t - \bar{r})^2,$$

  with $\bar{r}$ the mean of r_t on those days. The other 216 days of $\mathcal{T}_{\mathrm{oos}}$ enter forecast accuracy only. The risk-free return over the half-hour holding period is negligible and is not subtracted.

  E17. Decision: recompute the printed ratios on these days.
  Problem: the formula sums over the 614 days after the warmup, and every printed Sharpe ratio was computed on all 866 option days.
  The sizing bullet and this formula both score the sign strategy on the days $t > 252$, so the sign strategy and the sizing rules share one sample. The ratios printed in the results and in the economic-value list were computed on every day of $\mathcal{T}_{\mathrm{opt}}$, and that computation was not repeated on the post-warmup days.

- **Mean return and its HAC t-statistic:** the statistic of Appendix H on r_t, with $|\mathcal{O}| = T_{\mathrm{opt}}$ and $q_{\mathrm{NW}} = 14$.
- **Hit rate** $\frac{1}{T_{\mathrm{opt}}} \sum_{t \in \mathcal{T}_{\mathrm{opt}}} \mathbf{1}\{r_t > 0\}$; number of long positions $\#\{t : q_t > 0\}$.

- **Gain over the short-straddle benchmark** (or over any benchmark strategy B), with the daily return differential $\Delta_t = r_t^A - r_t^B$:

$$\Delta\mathrm{SR} = \mathrm{SR}(r^A) - \mathrm{SR}(r^B).$$

  The HAC t-statistic of Appendix H on Δ_t tests the mean return differential, not the equality of the two Sharpe ratios (Ledoit and Wolf, 2008).
- **Elementary score** of the Murphy diagram (Ehm et al., 2016) at the implied variance, the threshold used below:

$$\mathrm{ES}_{1,t} = \Big|\frac{\mathrm{RV}_t}{\mathrm{IV}_t^2} - 1\Big| \; \mathbf{1}\{\min(\hat{\sigma}_t^2, \mathrm{RV}_t) \le \mathrm{IV}_t^2 < \max(\hat{\sigma}_t^2, \mathrm{RV}_t)\},$$

  with mean $\overline{\mathrm{ES}_1}$ over $\mathcal{T}_{\mathrm{opt}}$.
- **The score $\mathrm{ES}_{1,t}$ is positive** only when $\hat{\sigma}_t^2$ and RV_t lie on opposite sides of the implied variance IV_t^2.
- **Compounding:** the premium bought or sold each day is a fixed fraction $f = 3\%$ of current wealth, so the terminal portfolio value, from $V_0 = 1$, and the annualized log growth rate are

$$V_{T_{\mathrm{opt}}} = \prod_{t \in \mathcal{T}_{\mathrm{opt}}} (1 + f\, r_t), \qquad G = \frac{252}{T_{\mathrm{opt}}} \sum_{t \in \mathcal{T}_{\mathrm{opt}}} \log(1 + f\, r_t).$$

- At quoted bid/ask prices, r_t^{ba} replaces $r_t = q_t R_t$. Maximum drawdown $\max_t(1 - V_t/\max_{t' \le t} V_{t'})$, with V_t the portfolio value after day t.

7.3 Results

- **The unconditional short-straddle benchmark** ($q_t \equiv -1$; "Short straddle" in the tables): SR = 0.20 at mid-quotes and -0.27 at quoted bid/ask prices. Its mean return is not significantly different from zero (HAC $t = 0.43$), and it loses money in 3 of the 5 calendar years: selling the closing straddle earns no statistically detectable variance risk premium in the option-data subsample.
- **Sign strategies of the intraday models versus the short-straddle benchmark** (point estimates): $\Delta\mathrm{SR} > 0$ for all 12 intraday forecasts.
  - 3 of the gains are significant at the 5% level (HAC t-statistic on the daily return differential), all three on the full predictor set: LightGBM ($t = 2.05$), XGBoost ($t = 1.99$) and the random forest ($t = 2.13$).
- **Highest Sharpe ratios** (point estimates), all on the full predictor set:
  - elastic net: SR = 1.50 at mid-quotes and 1.01 at quoted bid/ask prices (HAC $t = 1.90$ on its gain over the short-straddle benchmark);
  - ridge: 1.44 at mid-quotes, 0.96 at quoted bid/ask prices;
  - LightGBM: 1.43 at mid-quotes, 0.93 at quoted bid/ask prices.
- **Full predictor set versus the HAR benchmark:** the full predictor set raises SR at mid-quotes in all 6 model classes, from 0.69–1.01 to 1.20–1.50.
  - ΔSR from +0.30 to +0.61 (median +0.51; elastic net +0.61, ridge +0.53, lasso +0.51);
  - the gain is economically large but not significant at the 5% level in any class (HAC t-statistic on the daily return differential).

E16. Decision: move "not significant" onto the mean return, or cite a Sharpe test.
Problem: this item says the Sharpe gain is not significant, and the HAC t in parentheses tests the mean of the return gap Δ_t. That is a different hypothesis.
This item says the full predictor set raises the Sharpe ratio of the sign strategy in all 6 classes, by +0.30 to +0.61, and that the gain is not significant in any class. The test in parentheses is a HAC t-statistic on the daily return gap $\Delta_t = r_t^A - r_t^B$, with Newey–West lag $q_{\mathrm{NW}} = 14$. Section 7.2 builds that t as a test of the mean of Δ_t, and says it does not test the equality of the two Sharpe ratios. The Sharpe gap $\mathrm{SR}(r^A) - \mathrm{SR}(r^B)$ divides each strategy's mean by its own standard deviation, so a test of the mean gap is a different statement from a test of the Sharpe gap. This item attaches "not significant" to the Sharpe gain.

- **Position sizing,** median SR at mid-quotes across the 12 intraday forecasts: sign strategy 0.90, robust clip 1.37, rank 1.40.
  - $\mathrm{SR}^{\mathrm{clip}} > \mathrm{SR}^{\mathrm{sign}}$ for 11 of the 12 forecasts (point estimates; no test).
- **The value is concentrated in a few days**:
  - without its 10 best days, the mean return differential over the short-straddle benchmark, $\bar{\Delta}$, is negative for all 12 intraday forecasts;
  - days on which the package expires worthless ($R_t = -1$, settlement pins) supply most of the return.
- **Statistical accuracy is a weak guide to economic value.** Spearman rank correlations across all 112 forecasts of the model set (both training schemes), on $\mathcal{T}_{\mathrm{opt}}$, between average loss and the Sharpe ratio (negative: the more accurate forecasts earn the higher SR):

$$\rho_S(\overline{\mathrm{QLIKE}}, \mathrm{SR}) = -0.20, \qquad \rho_S(\overline{\mathrm{ES}}_1, \mathrm{SR}) = -0.55.$$

- **Robustness** (Appendix F):
  - shift test, $q_t = \mathrm{sign}(\hat{y}^2_{t,c+j}\, s^2_{t,c} - \mathrm{IV}_t^2)$, with the forecast of the interval j half hours away from the closing interval ($j < 0$: earlier on day t; $j > 0$: on day $t+1$): the median SR ranges from 0.00 to 1.58 across j (a swing of ±0.79), within the ±1.06 95% sampling band of an annualized Sharpe ratio under i.i.d. returns, so differences of this size are sampling noise;
  - subsample splits around May 2022;
  - compounding, across the 112 forecasts of the model set: median terminal portfolio value $V_{T_{\mathrm{opt}}}$ of 4.4 at mid-quotes and 1.8 at quoted bid/ask prices; maximum drawdowns of up to 64% and 78%, respectively.

7.4 A final refinement: training on the last 30 minutes

- **The scheme** (the closing-interval model, $\mathcal{I} = \{c\}$): every model class of Section 5.2 is estimated on the last 30 minutes of each trading day only, the closing interval $i = c$ (15:30–16:00 ET), as the benchmark forecast is.
  - One observation per trading day, so a period is a trading day and $\mathcal{W}_t = \{t-W, \ldots, t-1\}$ holds $W = 2{,}000$ trading days (Appendix B.2). Standardization of the series averages on this window: Appendix B.3.
  - Predictor sets, evaluation span and trading rules as before; accuracy in Table 2, on the same days as Table 1.
- **Training on the last 30 minutes lowers average QLIKE further,** in every model class and on both predictor sets: the intraday model's average QLIKE exceeds that of the closing-interval model of the same class, $\Delta\% = 100(\bar{L}^{\mathrm{pool}}/\bar{L}^{\mathrm{close}} - 1) > 0$, with $\bar{L}^{\mathrm{pool}}$ and $\bar{L}^{\mathrm{close}}$ the average losses of $f_{m,\mathcal{X},\{1,\ldots,M\}}$ and $f_{m,\mathcal{X},\{c\}}$:

  - linear models: +10.0 to +10.7% on the HAR benchmark (DM from 3.09 to 3.22, significant at the 5% level) and +5.3 to +11.1% on the full predictor set (significant for the lasso only);
  - trees: +18.5 to +21.7% on the HAR benchmark and +7.5 to +9.1% on the full predictor set, each significant at the 5% level.
- **The feature result holds:** the full predictor set lowers average QLIKE in all 6 closing-interval model classes, $\Delta\%$ from -4.1 to -0.1, none significantly at the 5% level (lasso: $\Delta\% = -4.0$, DM $= -1.91$, $p = 0.056$).
  - Across the 51 predictor-set pairs of the model set (45 tree and 3 linear specifications trained on the last 30 minutes, 3 intraday), it lowers average QLIKE in 35, 4 of them significantly.
- **The most accurate forecasts of the paper** are closing-interval models on the full predictor set: average QLIKE lasso 0.0992, elastic net 0.1014, ridge 0.1031, XGBoost 0.1031; the benchmark forecast ($\overline{\text{QLIKE}} = 0.1033$) ranks 8 of 116.
  - Against the benchmark forecast, the lasso changes average QLIKE by -4.0% (DM $= -1.91$, $p = 0.056$); none of the four is significantly more accurate at the 5% level.
  - Of the 111 other forecasts of the model set, 7 have a lower average QLIKE than the benchmark forecast, 0 of them significantly, and 56 are significantly less accurate.
  - Trees versus the closing-interval ridge on the same predictors: the tree has the higher average QLIKE in 48 of 48 comparisons, 23 of them significantly.
- **Squared-error loss disagrees:** the full predictor set raises the MSE in 49 of the 51 pairs of the model set, 0 of them significantly.
  - 19 March 2020, a day outside the option-data subsample, carries 28–39% of the closing-interval trees' total squared error over $\mathcal{T}_{\text{oos}}$ but at most 0.9% of their total QLIKE loss.
  - Five days in March 2020 carry 73–87% of the squared error of the benchmark forecast and of the closing-interval lasso and XGBoost on the full predictor set; excluding February–April 2020, that lasso has a lower MSE than the benchmark forecast.
- **Economic value** (sign strategy on $\mathcal{T}_{\text{opt}}$; point estimates):
  - The highest Sharpe ratio of the model set: the elastic net on the full predictor set, SR $= 1.92$ at mid-quotes and 1.43 at quoted bid/ask prices, its gain over the short-straddle benchmark significant at the 5% level (HAC $t = 2.56$); the lasso on the full predictor set: 1.88 and 1.40. Both exceed every intraday forecast (SR ≤ 1.50 at mid-quotes).
  - LightGBM, the best of its specifications by SR on $\mathcal{T}_{\text{opt}}$ (an ex post selection): 1.81.
  - Full predictor set minus HAR benchmark: ΔSR $= +0.55$ (elastic net) and $+0.50$ (lasso), neither significant at the 5% level; across the 51 pairs of the model set, $\Delta\text{SR} > 0$ in 50 (median +0.54), 4 of them significantly.
  - Against the short-straddle benchmark, $\Delta\text{SR} > 0$ for 112 of the 112 forecasts of the model set; 20 of the gains are significant at the 5% level, all of them with the full predictor set.
  - None of the differences in SR between the intraday and the closing-interval model of the same class is significant at the 5% level (HAC t-statistic on the daily return differential, $q_{\text{NW}} = 14$).

Table 2: Forecast accuracy when training on the last 30 minutes

	Last 30 minutes		
Model and predictor set	MSE (10^{-10})	QLIKE	R^2_{OOS} (%)
Panel A: Linear models			
OLS (benchmark forecast)			
HAR benchmark	4.15	0.1033	78.2
Ridge			
HAR benchmark	4.15	0.1033	78.2
All features	7.30	0.1031	61.7
Lasso			
HAR benchmark	4.15	0.1033	78.2
All features	5.94	0.0992	68.8
Elastic net			
HAR benchmark	4.15	0.1032	78.2
All features	6.89	0.1014	63.9
Panel B: Tree ensembles			
LightGBM			
HAR benchmark	5.75	0.1094	69.8
All features	17.19	0.1049	9.8
XGBoost			
HAR benchmark	5.27	0.1076	72.4
All features	20.94	0.1031	−9.9
Random forest			
HAR benchmark	4.62	0.1068	75.8
All features	14.15	0.1060	25.8

Notes: Rolling forecasts from 3 January 2020 to 30 April 2024, every forecast made with a model estimated on its trailing window only, so every forecast is out of sample; each is scored against the actual realized variance of the closing half hour, RV_t, on the $T_{\text{oos}} = 1{,}082$ trading days on which every forecast of both training schemes exists, early closes dropped (Section 5.3). The span starts with the intraday linear forecasts, on the first trading day after the pre-sample on which the penalties were selected (6 January 2016 to 2 January 2020). The MSE entries are in units of 10^{-10}: the printed number times 10^{-10} is the mean squared error $T_{\text{oos}}^{-1}\sum_t(\text{RV}_t - \hat{\sigma}_t^2)^2$. QLIKE is the mean QLIKE loss. $R^2_{\text{OOS}} = 1 - \sum_t(\text{RV}_t - \hat{\sigma}_t^2)^2 / \sum_t(\text{RV}_t - \overline{\text{RV}}_{t-1})^2$, in percent, where the benchmark $\overline{\text{RV}}_{t-1}$ is the historical mean of the closing-half-hour realized variance over all trading days from 5 January 1998 to day $t-1$ (Campbell and Thompson, 2008). All features is the full predictor set of Section 4. Each model is estimated on the last 30 minutes of each trading day only (one observation per day); the OLS on the HAR benchmark is the benchmark forecast. Trees use fixed hyperparameters and are re-estimated daily (Section B.2). The intraday models on the same days are in Table 1.

8 Conclusion

- **Exogenous features lower the QLIKE of the intraday forecast.** Against the HAR benchmark, the full predictor set lowers average QLIKE in all 6 intraday classes, 4 of the reductions significant at the 5% level (Section 6). The tree ensembles have the largest reductions. Among the linear models, the elastic net's reduction is significant.
- **Squared error does not confirm the QLIKE reductions.** The full set raises the MSE in 3 of the 6 classes. No squared-error difference is significant.
- **The intraday forecasts remain less accurate than the daily HAR benchmark.** Every intraday forecast has a higher average QLIKE than the benchmark forecast, the daily OLS on the HAR benchmark. The full predictor set narrows that gap in every class. The linear models' gap is not significant at the 5% level; the trees' gap is (Section 6.4).
- **Training on the closing interval is a refinement.** Estimating each model on the last 30 minutes lowers average QLIKE further, and the full predictor set still lowers QLIKE in every class (Section 7.4).
- **The straddle reads the same forecast.** On the option-data subsample, the full predictor set raises the sign strategy's Sharpe ratio in every intraday class. Average QLIKE and that Sharpe ratio are weakly related (Section 7.3).
- **Limits.**
  - The accuracy comparison uses $T_{\text{oos}} = 1{,}082$ trading days, 3 January 2020 to 30 April 2024.
  - One index, and one scored interval, the closing half hour.
  - The straddle is held to expiration with no delta hedge, and trading at the quoted bid and ask lowers the Sharpe ratio (Section 7).
- **Left open.** Expirations after the evaluation span, a second market, and forecast origins earlier in the day.

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A Appendix structure

- **Implementation details** (Appendix B): samples and windows, the training schemes in practice, the tuned hyperparameters and the tuning procedures.
- **Statistical tests** (Appendix H): the Diebold–Mariano statistic, the Newey–West variance and the HAC t-statistic.
- **A. Data and predictors** (Appendix C):

  - sources and spans;
  - predictor dictionary: 47 exogenous series $\rightarrow$ 628 predictors in the full predictor set (Appendix Table 5);
  - missing and stale data;
  - removal of duplicate predictors; the window filter (constant and duplicate predictors dropped in each estimation window $\mathcal{W}_t$);
  - SPXW option chain; implied-volatility inversion;
  - FOMC calendar.
- **B. Hyperparameter selection by model** (Appendix D):
  - linear models: penalty strength λ, elastic-net mixing α; block-penalized ridge, penalty weights ω_j;
  - tree ensembles: fixed hyperparameters, Optuna budgets, selected trials; boosting rounds $b = 1, \ldots, B$, learning rate η;
  - re-selection frequency, on the estimation sample $\mathcal{W}_t^{\text{est}}$, gap $\mathcal{W}_t^{\text{gap}}$ and validation sample $\mathcal{W}_t^{\text{val}}$; re-estimation frequency;
- **C. Robustness, statistical** (Appendix E):
  - full model table, tree ensembles included;
  - Diebold–Mariano and HAC conventions (DM with the Harvey–Leybourne–Newbold correction; Newey–West lag truncation q_{NW});
  - model confidence set, superior predictive ability, StepM;
  - regime and break tests;
  - closing-interval vs intraday models;
  - remaining variance through the trading day.
- **D. Robustness, trading** (Appendix F):
  - date-shift test;
  - Murphy diagrams (elementary scores); subsample splits;
  - P&L decomposition;
  - compounding;
  - position sizing;
  - strategy variations; other forecast origins; month-ends.
- **E. Reproducibility** (Appendix G):
  - software lock files;
  - hashes of the design matrices;
  - computing environment;
  - CPU classes;
  - compute time;
  - the generator script behind every reported number.

B Implementation details

B.1 Samples and windows

- **Estimation and evaluation samples:** Table 3.

E20. Decision: two window cells.
Problem: Panel B prints one intraday window, $W = 24{,}000$, and the ridge, lasso, and elastic-net QLIKE rows were fit on $W = 96{,}000$.
Panel B of Table 3 prints the intraday estimation sample as one row, and the window on that row is $W = 24{,}000$. Section B.2 and the note under the table give that window to the trees. Ridge, lasso, and elastic net are estimated on the longer window $W = 96{,}000$. The rolling-forecast row directly underneath is computed by subtracting 24,000 from the estimation-sample length, so the start date on that row is the first date at which a tree window is full. A linear window is full only after 96,000 intervals, and that origin is later. The QLIKE rows for ridge, lasso, and elastic net were computed from fits on the longer window. The cell they would be read against shows the shorter one.

- **Burn-in:** the first 3,125 intervals, the longest HAR horizon, are dropped (Section 3).
- **Diurnal adjustment** $s^2_{t,i}$ (Section 5.1): $n_{\mathrm{d}} = 20$ trading days, at least $\underline{n}_{\mathrm{d}} = 5$ of them; intervals with fewer have $s^2_{t,i} = 1$ and are not scored.
- **Winsorization** of the target: $n_{\mathrm{w}} = 240$ intervals, all intervals of the day pooled; the bounds are the empirical 1st and 99th percentiles, $\underline{Q}_{t,i} = Q_{0.01}(\mathcal{Y}_{t,i})$ and $\overline{Q}_{t,i} = Q_{0.99}(\mathcal{Y}_{t,i})$.
- **Hyperparameter-selection splits:** $n_{\mathrm{gap}} = 25$ and $n_{\mathrm{val}} = 125$ observations.

E21. Decision: label each length with its unit and which step uses it.
Problem: $n_{\mathrm{gap}} = 25$ and $n_{\mathrm{val}} = 125$ are called observations, which on the intraday grid are 30-minute intervals, and the tuning text and Table 3 use the same symbols as numbers of trading days.
This item sets the embargo n_{gap} equal to 25 and the validation length n_{val} equal to 125, and it calls both of them observations. On the intraday grid an observation is one 30-minute interval, so an observation count is 48 times a count of trading days. The linear tuning item, a different step, describes its pre-sample in trading days: four folds of 250 days, running from 6 January 2016 to 2 January 2020. The light re-tune then says it uses two folds of 125 days, which puts the same symbol 125 on a count of days. Table 3 prints that pre-sample as four folds of 250 days after a gap of 25 days. Read as observations, 125 is a number of half hours. Read as in the re-tune and the table, it is a number of trading days.

B.2 Training schemes in practice

- **Closing-interval scheme:** $W = 2{,}000$ trading days for every model class; re-estimated every trading day, except the tree specifications re-estimated every 10 trading days (fixed hyperparameters or random search).
- **Intraday scheme** ($\mathcal{I} = \{1, \ldots, M\}$), the paper's training scheme: estimated on all $M = 48$ intervals of each day, overnight included; one observation per interval, so a period is an intraday interval and $\mathcal{W}_{t,i} = \{(t, i-W), \ldots, (t, i-1)\}$ holds the W intervals before (t, i). The full predictor set has 1,144 predictors (Section 4).
  - Only the forecasts of the closing interval $i = c$, made at the 15:30 ET forecast origin, are scored and traded.
  - Linear models: $W = 96{,}000$ intervals (2,000 trading days), re-estimated every interval.
  - Trees: $W = 24{,}000$ intervals; one fixed hyperparameter configuration per tree method, never re-selected, re-estimated every interval.

B.3 Closing-interval standardization

The closing-interval model of Section 7.4 standardizes the averages of series values on its own window $\mathcal{W}_t$ (Section B.2). Rows t' are trading days ($i = c$ suppressed):

$$z^{(k)}_t = \frac{\bar{u}^{(k)}_t - \mathrm{med}_{t' \in \mathcal{W}_t} \bar{u}^{(k)}_{t'}}{\max\{\mathrm{IQR}^{(k)}_t,\ 0.01 \max_{t' \le t} \mathrm{IQR}^{(k)}_{t'},\ \mathrm{IQR}^{*(k)}_t\}}, \qquad \mathrm{IQR}^{(k)}_t = \mathrm{IQR}\{\bar{u}^{(k)}_{t'} : t' \in \mathcal{W}_t\}.$$

- $\mathrm{IQR}^{*(k)}_t$: IQR of $\bar{u}^{(k)}_{t'}$, $t' < t$, over rows whose k-interval averaging window is fully observed.

- Column whose mode accounts for at least 20% of rows: $\mathrm{IQR}_t^{*(k)}$ also excludes rows at the mode.
- $\mathrm{IQR}_t^{*(k)} = 1$ until 8 such rows exist.
- $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$: not standardized. HAR components, calendar dummies: $\mathrm{IQR}_t^{*(k)} = 0$.
- $t < W$: the first window's statistics.

B.4 Tuned hyperparameters

- **Linear penalties,** chosen by hyperparameter selection on a pre-sample (Section B.5), one grid value per decade (Table 4):

$$\lambda^{\text{ridge}} \in \{10^{-2}, \ldots, 10^{3}\}, \qquad \lambda^{\text{lasso}}, \lambda^{\text{enet}} \in \{10^{-6}, \ldots, 10^{-2}\}.$$

  For lasso and elastic net these are the scikit-learn parameters (Section 5.2).
  - full predictor set: ridge 10^3, lasso and elastic net 10^{-2} (top edge);
  - HAR benchmark: the lower edge of each grid.

  E23. Decision: either the penalty stayed at the lower edge, or the four replacements need dates and values.
  Problem: the HAR-benchmark λ in the table is the pre-sample value. The penalty was replaced on 4 of 24 re-selections, and the four dates and the four new values are missing.
  This item says the penalty chosen for the HAR benchmark is the bottom of its grid, ridge at 10^{-2} and lasso and elastic net at 10^{-6}, and the penalty chosen for the full predictor set is the top of its grid, ridge at 10^3 and lasso and elastic net at 10^{-2}. The note under Table 4 describes a second step. At each re-selection a neighboring grid point may replace the incumbent, and that replacement happened 4 times out of 24: once for ridge, once for lasso, and twice for elastic net on the HAR benchmark, and on no occasion for any of the three on the full set. Because the full set never moved, the full-set column of the table is the penalty in force at every forecast origin. Because the HAR benchmark moved four times, the HAR-benchmark column is only the value chosen on the pre-sample. The date of each of the four replacements, and the neighbor that took over, are not stated, so the penalty behind a HAR-benchmark forecast on a given day cannot be read off the table.

- **Trees,** fixed hyperparameters from a search on a 1998–2002 pre-sample (Table 4):
  - LightGBM and XGBoost: the best trial of an Optuna search (Akiba et al., 2019);
  - random forest: trial 255 of 400 ($B = 485$ trees, maximum depth 20, share of predictors tried at each split 0.90, bootstrap sample share 0.41);
  - leaf minimums are rescaled to the 2,000-observation window: $n_{\min} \leftarrow n_{\min} \cdot 2{,}000/24{,}000$, the intraday trees' window (Section B.2).

  E24. Decision: the parenthetical is the closing-interval leaf size.
  Problem: this sentence calls the rescaled $n_{\min}$ the intraday trees' window. The intraday fits use the unscaled $n_{\min}$, selected on 24,000 intervals. The number in parentheses is the closing-interval leaf size. The leaf minimum $n_{\min}$ was selected on a window of 24,000 intervals. This item rescales it by the ratio of window lengths, $n_{\min} \leftarrow n_{\min} \cdot 2{,}000/24{,}000$, and the sentence calls that product the window of the intraday trees. The note under Table 4 assigns the two numbers the other way around. The figure outside the parentheses is the minimum selected on the 24,000-interval window, and the figure inside the parentheses is that minimum rescaled to a window of 2,000 days. Section B.2 fits the intraday trees on 24,000 intervals, so those fits use the outside number, the one that was actually selected. The closing-interval trees are re-estimated every 10 days on a window of 2,000 days, and those are the fits that use the parenthetical.

B.5 Tuning procedures

- **Linear:**

  - hyperparameter selection on a pre-sample of four folds of 250 trading days (6 January 2016 to 2 January 2020), on validation MSE:

$$\hat{\lambda} = \arg\min_{\lambda} \frac{1}{4} \sum_{v=1}^{4} \mathrm{MSE}_v^{\mathrm{val}}(\lambda),$$

    with $\mathrm{MSE}_v^{\mathrm{val}}(\lambda)$ the mean squared error of $\tilde{y}$ on the validation sample of fold v;
  - a light re-tune at each re-tuning date (every 250 trading days): the grid neighbors $\mathcal{C} \subseteq \{\lambda^-, \lambda^+\}$ (one decade below and above, where on the grid) challenge the incumbent λ;
  - the best challenger λ' (lowest validation MSE) replaces it only if the Diebold–Mariano statistic DM of Appendix H, on the paired squared errors, is significantly negative:

$$\mathrm{DM}(\{e_{t'}^2(\lambda') - e_{t'}^2(\lambda)\}_{t'}) < -z_{0.05/|\mathcal{C}|}, \qquad z_{0.05/|\mathcal{C}|} = \Phi^{-1}(1 - 0.05/|\mathcal{C}|);$$

  - $e_{t'}(\lambda) = \tilde{y}_{t'} - \hat{y}_{t'}(\lambda)$, t' over two 125-day validation folds; Φ: the standard normal distribution function.

- **Trees,** in increasing order of tuning effort:
  - fixed hyperparameters, re-estimated every 10 trading days or daily;
  - random search every 250 trading days;
  - Optuna (Akiba et al., 2019): 50 trials, re-tuned every 1, 5, 25 or 250 trading days, best of the first 10, 25 or 50 trials.

    E22. Decision: say which rung is in Table 1, and park the others.
    Problem: the schemes section, the three-rung list in this subsection, and the draft or preliminary flag on the full-set tree rows each claim a tree cell in Table 1, and no cell is tied to one of them.
    A tree cell in Table 1 is the loss of one fitting procedure, and three procedures are written down as if each of them were that procedure. Section B.2 gives each tree method a single configuration, chosen once and never re-selected, and re-estimates it every interval on a window of $W = 24{,}000$. The list in this subsection gives three grades of tuning instead: fixed hyperparameters re-estimated every 10 trading days or every day, a fresh random search every 250 trading days, and an Optuna search of 50 trials re-tuned every 1, 5, 25, or 250 trading days. Section 6.1 adds that the full-set LightGBM and XGBoost rows are an earlier draft and the random forest is preliminary, so those rows need not be any of the procedures just listed. Until one procedure is marked as the source of a given QLIKE cell, the cell has no window, no refit cadence, and no search attached to it.

- **Every selection uses only data available at the forecast origin;** no forecast is scored on a day that entered the selection of its own hyperparameters, and the evaluation span $\mathcal{T}_{\mathrm{oos}}$ starts after the pre-sample.

C Data and predictors

- **Sources and spans.** Eight groups, in Section 3. Closing-interval models use all 47 exogenous series. Intraday models use the 43 series in every group except FOMC.
- **Predictor dictionary.** Appendix Table 5 defines each series. Trailing averages, activity indicators, and calendar columns are added in Section 4.
- **Missing and stale values.** For an exogenous series u, the availability indicator $a_{t,i} = \mathbf{1}\{u \text{ observed at } (t,i)\}$ is recorded before any filling (Section 3). The series is then forward-filled with no staleness limit. On every interval with $a_{t,i} = 0$ the stored regressor is the imputation of Section 4, step 4: the median of the series' observed adjusted values on earlier retained intervals, or the adjusted value on its first observed interval when no earlier observation exists.

- **Duplicate and constant predictors.** At each estimation the window screen of Section 4 drops a column that is constant on $\mathcal{W}_{t,i}$ or an exact bitwise copy of an earlier kept column. The first copy is kept. On the closing-interval model with the full predictor set the screen keeps a median of 369 of the 628 columns.
- **SPXW chain.** Quotes on S&P 500 Weeklys that expire the same day, every 30 minutes: bid, ask, mid-quote, underlying level, implied volatility, and delta. The file covers 1,291 expirations, 2020–2025. The straddle in Section 7.1 uses the 15:30 quotes. A day is dropped when the wider of the two strikes is more than 10 index points from S_t, or when fewer than 10 contracts are quoted at that stamp. S_t is the 15:30 underlying print on the chain, and the official close is the S&P 500 close.
- **Implied-volatility inversion.** The vendor quote on each leg is a volatility per hour, from a geometric bisection. A solve that does not converge returns a bracket node of that bisection; those nodes are set to missing before the two legs are averaged, and a missing leg makes the package's implied variance missing. The 16:00 stamp is not used. The conversion from the hourly quote to the closing half hour is the display in Section 7.1.
- **FOMC calendar.** The four scheduled-announcement series of Section 3, on closing-interval models only. The schedule is known before the interval.

D Hyperparameter selection by model

- **Penalty parameters.** Ridge, lasso, and elastic net minimize the raw sum of squares plus the penalty in Section 5.2. λ is the coefficient on that penalty for ridge. For lasso and elastic net the printed grid is the scikit-learn parameter, and the coefficient in the sum is $2n$ times the printed value. Lasso is $\alpha = 1$. Elastic net is $\alpha = 0.5$. The block weights ω_j are 0.01 on the HAR benchmark, 1 on an additional predictor, and 0 on the intercept. The HAR benchmark fitted alone uses $\omega_j = 1$ on every predictor.
- **Where λ is chosen.** One grid value per decade, Table 4. The pre-sample is four folds of 250 trading days, 6 January 2016 to 2 January 2020, and the criterion is validation MSE (Section B.5). A light re-tune every 250 trading days lets the neighboring grid points challenge the incumbent. The best challenger replaces it only when the Diebold–Mariano statistic on the paired squared errors is significantly negative. The full predictor set stays at the top of its grid. The HAR-benchmark column of the table is the pre-sample value; the four replacements are note E23.
- **Trees.** LightGBM and XGBoost take the best trial of an Optuna search on a 1998–2002 pre-sample. The random forest takes trial 255 of 400. The selected configuration and the leaf minimum are in Section B.4. Which of the three tuning rungs in Section B.5 produced a tree cell of Table 1 is note E22.
- **What may enter a choice.** A selection uses only data available at the forecast origin. The evaluation span starts after the pre-sample.

E Robustness of the statistical results

- **Reported comparisons.** Forecast accuracy on $\mathcal{T}_{\text{oos}}$ is Table 1. The Diebold–Mariano statistic, with the Harvey–Leybourne–Newbold correction, and the Newey–West variance and HAC t-statistic are Appendix H.
- **Model confidence set, superior predictive ability, and StepM.** These three procedures

are named in that appendix for the comparison of predictor sets. No table reports an MCS membership, an SPA p-value, or a StepM rejection (note E14).

- **Regime and break tests.** Regime tests and break tests have no reported output.
- **Closing-interval against intraday.** Both schemes forecast the same closing half hour. The comparison of their average QLIKE is in Section 6. A forecast of the variance remaining from an earlier origin to the close is not reported.

F Robustness of the trade

- **Date-shift test.** The sign strategy is re-scored with the forecast of the interval j half hours away from the closing interval, placed on the closing interval's diurnal adjustment:

$$q_t = \text{sign}(\hat{y}^2_{t,c+j}\, s^2_{t,c} - \text{IV}^2_t).$$

  $j < 0$ reads an earlier interval of day t. $j > 0$ reads day $t+1$. Across j, the median annualized Sharpe ratio runs from 0.00 to 1.58 (a swing of ± 0.79), inside the ± 1.06 95% sampling band of an annualized Sharpe ratio under i.i.d. returns.
- **Murphy diagram.** The elementary score at the implied variance is $\text{ES}_{1,t}$ in Section 7.2. Its rank correlation with the Sharpe ratio, across the 112 forecasts of the model set, is in Section 7.3. The subsample split is around May 2022.
- **Profit and loss.** The reported return is R_t in Section 7.1. A decomposition of that return into separate variance and direction components is not reported.
- **Compounding.** Wealth starts at 1 and the premium each day is the fraction $f = 3\%$ in Section 7.2. Across the 112 forecasts, the median terminal value is 4.4 at mid-quotes and 1.8 at quoted bid and ask prices, and the maximum drawdowns reach 64% and 78%.
- **Position sizing.** The sign strategy and the three sized rules are in Section 7.1. All four are scored on the 614 days after the warmup of 252 days.
- **Other origins.** The date-shift test is the check at origins other than 15:30. A separate month-end strategy is not reported.

G Reproducibility

- **Generator of each reported number.**
  - `writeup/make_rehaul_data_methods.py`: samples, windows, hyperparameter tables, and the predictor dictionary.
  - `writeup/make_rehaul_oos_table.py`: Table 1 and the last-30-minutes split.
  - `writeup/make_rehaul_results.py`: the accuracy macros in Section 6.
  - `writeup/make_rehaul_economics.py`: the straddle macros in Section 7.
  - `writeup/make_rehaul_fi.py`: the importance macros in Section 6.5.
  - `writeup/make_rehaul_appendix.py`: the appendix-table macros.
- **Software list.** `requirements.txt` at the repository root names the packages and a version range for each. It is not a pinned lock file, and the draft does not record a second lock file.
- **Design matrices, CPU class, and compute time.** The generators read the forecasts and tables already on disk. The draft does not record a hash of the design matrix behind Table 1, the CPU class of those fits, or their compute time.

Table 3: Estimation and evaluation samples.

Sample	Start	End	Obs.	Window W
Intraday grid: 30-minute intervals with an observed $\mathrm{RV}_{t,i}$, outside the weekend break	1998-01-05	2024-04-30	303,442	–
Panel A: Closing-interval models (trading days)				
Closing-interval observations, after the burn-in	2010-07-12	2024-04-30	3,469	2,000
Rolling-window forecasts	2018-06-25	2024-04-30	1,469	–
Hyperparameter-selection pre-sample (linear models)	2004-04-13	2020-01-02	3,947	2,000
Validation samples (4×250 days)	2016-01-06	2020-01-02	1,000	2,000
Panel B: Intraday models (30-minute intervals)				
Estimation sample, after the 3,125-interval burn-in	1998-06-04	2024-04-30	300,317	24,000
Rolling-window forecasts	2001-07-13	2024-04-30	276,317	–
Tree hyperparameter pre-sample (before 2003)	1998-06-04	2002-12-31	36,285	24,000
Panel C: Evaluation samples (trading days)				
Evaluation span (T_{oos}): trading days on which every forecast exists, less early-close days (6)	2020-01-03	2024-04-30	1,082	–
Option-data subsample (T_{opt}): SPXW expiration days (872) less early-close days (6)	2020-01-03	2024-04-30	866	–
Later SPXW expiration days, held out (413 full trading days)	2024-05-01	2025-12-31	419	–

Notes: Dates are interval-end dates (ET); Obs. and the rolling estimation window W are in the units of each panel. The burn-in, 3,125 intervals, is the longest HAR horizon and is applied to each series as built (the closing-interval series, the intraday grid and, for the linear pre-sample, the series of the two intervals of the last hour), so the closing-interval sample starts in 2010 and the pre-sample in 2004. The pooled linear models use $W = 96{,}000$ intervals (2,000 trading days); the tree pre-sample is re-estimated every 250 intervals. Each validation sample follows a 25-day gap and a 2,000-day estimation sample (Section B.1). Full trading days: expiration days that are not early-close days. Panel C: the evaluation span holds every trading day on which all rolling forecasts exist (Section 5.3); the straddle economics use its option-data subsample (Section 7).

Table 4: Hyperparameters: penalized regressions and tree ensembles.

Model	Hyperparameter	Search space	Value: HAR benchmark	Value: Full predictor set
Panel A: Penalized linear models, selected on the pre-sample				
OLS	–	–	–	–
Ridge	Penalty, λ	$\{10^{-2}, 10^{-1}, \ldots, 10^{3}\}$ (6)	10^{-2}	10^{3}
Lasso	Penalty, λ	$\{10^{-6}, 10^{-5}, \ldots, 10^{-2}\}$ (5)	10^{-6}	10^{-2}
Elastic net	Penalty, λ	$\{10^{-6}, 10^{-5}, \ldots, 10^{-2}\}$ (5)	10^{-6}	10^{-2}
	Mixing weight, α	fixed	0.5	0.5
All three	Penalty weight on the HAR benchmark, $\underline{\omega}$	fixed	1	0.01
Panel B: Tree ensembles, fixed hyperparameters (selected on the 1998–2002 pre-sample)				
LightGBM	Leaves per tree, J	[15, 255] (log)	20	
	Learning rate, η	[0.01, 0.3] (log)	0.013	
	Boosting rounds, B	[50, 500]	392	
	Minimum leaf size, $n_{\min}$	[20, 500] (log)	98 (8)	
	Share of predictors per tree	[0.5, 1]	0.70	
	Share of observations per tree	[0.5, 1]	0.56	
	L_1 penalty on leaf weights, λ_1^{tree}	$[10^{-8}, 10]$ (log)	10^{-8}	
	L_2 penalty on leaf weights, λ_2^{tree}	$[10^{-8}, 10]$ (log)	0.016	
XGBoost	Maximum depth	[3, 10]	5	
	Learning rate, η	[0.01, 0.3] (log)	0.0123	
	Boosting rounds, B	[50, 500]	367	
	Minimum leaf size, $n_{\min}$	[1, 100] (log)	50.2 (4.18)	
	Share of observations per tree	[0.5, 1]	0.66	
	Share of predictors per tree	[0.5, 1]	0.65	
	L_1 penalty on leaf weights, λ_1^{tree}	$[10^{-8}, 10]$ (log)	0.0215	
	L_2 penalty on leaf weights, λ_2^{tree}	$[10^{-8}, 10]$ (log)	1.2×10^{-5}	
	Penalty per leaf, γ	$[10^{-8}, 1]$ (log)	1.5×10^{-6}	
Random forest	Number of trees, N	[50, 500]	485	
	Maximum depth	[2, 32] (log)	20	
	Share of predictors per split	[0.02, 1] (log)	0.90	
	Bootstrap sample share	[0.05, 1] (log)	0.41	
	Minimum leaf size, $n_{\min}$	[1, 1000] (log)	12 (1)	

Notes: Panel A: Section 5.2 states the scale of λ. One grid value per decade (grid size in parentheses), selected by mean validation MSE over the 4 folds of the pre-sample (Table 3). At each of the 4 re-selection points a grid neighbor may replace it (Section B.5); this happened 4 times in 24 re-selections (ridge 1 and 0, lasso 1 and 0, elastic net 2 and 0, for the HAR benchmark and the full predictor set). Panel B: the best trial of an Optuna search on the intraday pre-sample (24,000-interval window; random forest: trial 255 of 400), ranked by MSE and then held fixed for both predictor sets; (log): searched on a logarithmic scale. $n_{\min}$ was selected on 24,000 intervals and is rescaled to the 2,000-day window (rescaled value in parentheses). Implementations: LightGBM (Ke et al., 2017) with bagging at every round, XGBoost (Chen and Guestrin, 2016) with histogram-based split finding and the scikit-learn random forest with bootstrap samples; random seeds are fixed.

Table 5: Predictor definitions.

Variable	Description	Source	Diurnal adjustment	Transform $g(v)$	Predictors, $k \in \mathcal{K}$
Realized moments (7; front-month E-mini, one-minute log-price changes)					
r	Sum of one-minute changes in the log price of the front-month contract, $r_{t,i} = \sum_k r_{t,i,k}$	E-mini S&P 500 futures	s.d.	v	$u^{(k)}$, $\bar{a}^{(k)}$
RAV	Realized absolute variation, $\sum_k \lvert r_{t,i,k} \rvert$	E-mini S&P 500 futures	mean	v	$u^{(k)}$, $\bar{a}^{(k)}$
RSK	Realized skewness, unnormalized, $\sum_k r_{t,i,k}^3$	E-mini S&P 500 futures	s.d.	$v^{1/3}$	$u^{(k)}$, $\bar{a}^{(k)}$
RQ	Realized quarticity, unscaled, $\sum_k r_{t,i,k}^4$	E-mini S&P 500 futures	mean	$\text{sign}(v)\,\lvert v \rvert^{1/4}$	$u^{(k)}$, $\bar{a}^{(k)}$
RS^+	Positive realized semivariance, $\sum_k r_{t,i,k}^2 \mathbf{1}\{r_{t,i,k} > 0\}$	E-mini S&P 500 futures	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
BV	Bipower variation, $\text{BV}_{t,i} = \sum_{k \geq 2} \lvert r_{t,i,k} \rvert \, \lvert r_{t,i,k-1} \rvert$	E-mini S&P 500 futures	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
AC	First-order realized autocovariance, $\text{AC}_{t,i} = \sum_{k \geq 2} r_{t,i,k}\, r_{t,i,k-1}$	E-mini S&P 500 futures	s.d.	$\text{sign}(v)\,\lvert v \rvert^{1/2}$	$u^{(k)}$, $\bar{a}^{(k)}$
Liquidity (14)					
Volume	Number of front-month contracts traded in the half hour	E-mini S&P 500 futures	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
N	Number of non-missing one-minute intervals in the half hour, $N_{t,i} \leq 30$; mode 30 on 72% of intervals	E-mini S&P 500 futures	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$
TO^{EW}	Equal-weighted turnover, shares traded in the half hour divided by the previous day's shares outstanding	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\text{TO}^{\text{EW}}_{\text{buy}}$	Equal-weighted buyer-initiated turnover, the same ratio using estimated buy orders	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\text{TO}^{\text{EW}}_{\text{sell}}$	Equal-weighted seller-initiated turnover, the same ratio using estimated sell orders	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\text{EffSpread}^{\text{EW}}$	Equal-weighted effective spread, the mean percent difference between the trade price and the preceding midquote	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\text{Spread}^{\text{EW}}$	Equal-weighted quoted spread, the mean percent difference between the bid and the ask	Constituents	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
TO^{VW}	Value-weighted turnover, shares traded in the half hour divided by the previous day's shares outstanding	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$

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Table 5 (continued)

Variable	Description	Source	Diurnal adjustment	Transform $g(v)$	Predictors, $k \in \mathcal{K}$
$\mathrm{TO}_{\mathrm{buy}}^{\mathrm{VW}}$	Value-weighted buyer-initiated turnover, the same ratio using estimated buy orders	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{TO}_{\mathrm{sell}}^{\mathrm{VW}}$	Value-weighted seller-initiated turnover, the same ratio using estimated sell orders	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{EffSpread}^{\mathrm{VW}}$	Value-weighted effective spread, the mean percent difference between the trade price and the preceding midquote	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{Spread}^{\mathrm{VW}}$	Value-weighted quoted spread, the mean percent difference between the bid and the ask	Constituents	s.d.	v	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{OFI}^{\mathrm{EW}}$	Equal-weighted order-flow imbalance, (Buy − Sell)/(Buy + Sell) of turnover	Constituents	s.d.	v	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{OFI}^{\mathrm{VW}}$	Value-weighted order-flow imbalance, (Buy − Sell)/(Buy + Sell) of turnover	Constituents	s.d.	v	$u^{(k)}$, $\bar{a}^{(k)}$
Equal-weighted market (6; simple averages of S&P 500 constituents)					
$\mathrm{RV}^{\mathrm{EW}}$	Equal-weighted realized variance of 5-minute log returns	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{RSK}^{\mathrm{EW}}$	Equal-weighted realized skewness of 5-minute log returns, unnormalized	Constituents	s.d.	$v^{1/3}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{RQ}^{\mathrm{EW}}$	Equal-weighted realized quarticity of 5-minute log returns, unscaled	Constituents	mean	$\mathrm{sign}(v)\,\lvert v\rvert^{1/4}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{RAV}^{\mathrm{EW}}$	Equal-weighted realized absolute variation of 5-minute log returns	Constituents	mean	v	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{BV}^{\mathrm{EW}}$	Equal-weighted bipower variation of 5-minute log returns	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{RS}^{+,\mathrm{EW}}$	Equal-weighted positive realized semivariance of 5-minute log returns	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
Value-weighted market (6; previous day's market capitalization)					
$\mathrm{RV}^{\mathrm{VW}}$	Value-weighted realized variance of 5-minute log returns	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{RSK}^{\mathrm{VW}}$	Value-weighted realized skewness of 5-minute log returns, unnormalized	Constituents	s.d.	$v^{1/3}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{RQ}^{\mathrm{VW}}$	Value-weighted realized quarticity of 5-minute log returns, unscaled	Constituents	mean	$\mathrm{sign}(v)\,\lvert v\rvert^{1/4}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{RAV}^{\mathrm{VW}}$	Value-weighted realized absolute variation of 5-minute log returns	Constituents	mean	v	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{BV}^{\mathrm{VW}}$	Value-weighted bipower variation of 5-minute log returns	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$

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Table 5 (continued)

Variable	Description	Source	Diurnal adjustment	Transform $g(v)$	Predictors, $k \in \mathcal{K}$
$\mathrm{RS}^{+,\mathrm{VW}}$	Value-weighted positive realized semivariance of 5-minute log returns	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
Sentiment (3)					
Attn	Number of StockTwits posts in the half hour	StockTwits	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
Sent	Score of tagged posts, 1 if all are bullish and -1 if all are bearish	StockTwits	none	v	$u^{(k)}$, $\bar{a}^{(k)}$
NSent	Number of posts in the half hour that carry a bullish or bearish tag	StockTwits	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
Implied volatility (3)					
VIX	VIX index (30-day S&P 500 implied volatility)	Cboe	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
VVIX	VVIX index (implied volatility of VIX)	Cboe	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
VIX3M	VIX3M index (3-month S&P 500 implied volatility)	Cboe	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
Volatility demand (4; net vega bought by non-market-makers)					
$\mathrm{VD}_{\mathrm{OC}}^{\mathrm{SPX}}$	Net vega bought by non-market-makers, S&P 500 index and top S&P 500 ETF options, trades that open or close a position; mode 0 on 59% of intervals	Cboe	none	$\operatorname{asinh}(u/\kappa)$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$
$\mathrm{VD}_{\mathrm{O}}^{\mathrm{SPX}}$	Net vega bought by non-market-makers, S&P 500 index and top S&P 500 ETF options, trades that open a position; mode 0 on 60% of intervals	Cboe	none	$\operatorname{asinh}(u/\kappa)$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$
$\mathrm{VD}_{\mathrm{OC}}^{\mathrm{All}}$	Net vega bought by non-market-makers, all traded options, trades that open or close a position; mode 0 on 59% of intervals	Cboe	none	$\operatorname{asinh}(u/\kappa)$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$
$\mathrm{VD}_{\mathrm{O}}^{\mathrm{All}}$	Net vega bought by non-market-makers, all traded options, trades that open a position; mode 0 on 60% of intervals	Cboe	none	$\operatorname{asinh}(u/\kappa)$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$
FOMC (4; closing-interval model only, scheduled calendar)					
$\mathrm{FOMC}^{\mathrm{rel}}$	FOMC announcement-interval dummy (14:30 ET); mode 0 on 98% of intervals	FOMC calendar	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$
$\mathrm{FOMC}^{\mathrm{day}}$	FOMC announcement-day dummy; mode 0 on 95% of intervals	FOMC calendar	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$
$\mathrm{FOMC}^{\mathrm{next}}$	$1/(1+n_{t,i}^{+})$, $n_{t,i}^{+}$ intervals to the next announcement	FOMC calendar	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$

$\text{FOMC}^{\text{last}}$	$1/(1+n^-_{t,i})$, $n^-_{t,i}$ intervals since the last announcement	FOMC calendar	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$

Notes: Each series u enters as $u^{(k)}$, its adjusted, transformed and winsorized value averaged over the previous k intervals, and as $\bar{a}^{(k)}(u)$, the share of those intervals on which it was observed; series with an activity indicator also enter as $\bar{\chi}^{(k)}(u)$, the share off their mode (the mode and its share of intervals are in the description). $\mathcal{K} = \{1, 5, 25, 125, 625, 3125\}$ in the closing-interval model and $\{1, 2, 4, \ldots, 2048\}$ in the intraday model, which is every group here except FOMC (Section 3). Diurnal adjustment: division by the trailing same-interval mean (mean) or standard deviation (s.d.); g: the variance-stabilizing map; the volatility-demand series take $\text{asinh}(u_{t,i}/\kappa_{t,i})$ in place of both (Section 4). A series is signed if it has any negative value: $\text{Spread}^{\text{VW}}$ has a single negative value (2018-11-12 11:00 ET) and so takes $g(v) = v$ and the s.d. adjustment, where $\text{Spread}^{\text{EW}}$ takes $\log v$ and the mean. Unnormalized and unscaled: without the factors $\sqrt{N_{t,i}}/\text{RV}_{t,i}^{3/2}$ (skewness), $N_{t,i}/3$ (quarticity) and $\pi/2$ (bipower variation).

H Statistical tests

- **Diebold–Mariano test** (Diebold and Mariano, 1995) on the daily QLIKE loss differential of two forecasts A and B, with the Harvey–Leybourne–Newbold small-sample correction for one-step-ahead forecasts (Harvey et al., 1997):

$$d_t = L_t^A - L_t^B, \qquad \mathrm{DM} = \sqrt{\frac{|\mathcal{O}|-1}{|\mathcal{O}|}}\,\frac{\bar{d}}{\sqrt{\hat{\Omega}/|\mathcal{O}|}}.$$

  - A, B: forecasts $f_{m,\mathcal{X},\mathcal{I}}$, indexed by the model m, the predictor set $\mathcal{X} \in \{\mathrm{HAR}, \mathrm{Full}\}$ and the intervals used in estimation $\mathcal{I} \in \{\{c\}, \{1,\dots,M\}\}$.
  - $\mathcal{O}$: the days on which the two series are compared, numbered $t = 1,\dots,|\mathcal{O}|$ in time order: $\mathcal{O} = \mathcal{T}_{\mathrm{oos}}$, $|\mathcal{T}_{\mathrm{oos}}| = T_{\mathrm{oos}}$, for forecast accuracy and the option-data subsample $\mathcal{T}_{\mathrm{opt}} \subset \mathcal{T}_{\mathrm{oos}}$ for the straddle returns (Section 3).
  - $\bar{d}$: mean loss differential over $\mathcal{O}$; two-sided p-value from the standard normal, $p = 2\{1 - \Phi(|\mathrm{DM}|)\}$, with Φ the standard normal distribution function.
  - $\mathrm{DM} < 0$ favors forecast A. "Significant" means significant at the 5% level ($p < 0.05$).
- **Newey–West variance** (Newey and West, 1987), Bartlett weights, lag truncation q_{NW}:

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{q_{\mathrm{NW}}}\Big(1 - \frac{j}{q_{\mathrm{NW}}+1}\Big)\hat{\gamma}_j, \qquad \hat{\gamma}_j = \frac{1}{|\mathcal{O}|}\sum_{t=j+1}^{|\mathcal{O}|}(d_t - \bar{d})(d_{t-j} - \bar{d}).$$

E13. Decision: change this display, or recompute the printed DM.
Problem: this display divides $\hat{\gamma}_j$ by $|\mathcal{O}|$, and every printed DM was computed by `dm_test`, which divides by $|\mathcal{O}| - j$.
The autocovariance $\hat{\gamma}_j$ in the display is the average of the products $(d_t - \bar{d})(d_{t-j} - \bar{d})$, and the divisor written in front of the sum is $|\mathcal{O}|$, the number of days in the comparison. There are only $|\mathcal{O}| - j$ such products, because the first j days have no lag-j partner. The function that produced the printed Diebold–Mariano statistics is `dm_test`, and it takes those products with `np.mean`, which divides by the length of the array it is given, $|\mathcal{O}| - j$. The HAC t on returns comes from `newey_west_t`, and that function divides by $|\mathcal{O}|$, so it matches this display. The accuracy-section DM numbers match `dm_test`. On $\mathcal{T}_{\mathrm{oos}}$ the lag runs from 1 through 6, and at each of those lags the two divisors differ.

  - DM: $q_{\mathrm{NW}} = \lfloor 4(|\mathcal{O}|/100)^{2/9} \rfloor$ (its value on $\mathcal{T}_{\mathrm{oos}}$: Experimental setup, Section 6.1).
- **HAC t-statistic of a mean daily return** (Newey–West, no HLN correction), for a daily return series r_t, either a single strategy's return (a test of its mean return) or the difference between two strategies' returns (a test of the mean return differential):

$$t = \frac{\bar{r}}{\sqrt{\hat{\Omega}_r/|\mathcal{O}|}},$$

  with $\hat{\Omega}_r$ the Newey–West variance above computed on r_t in place of d_t and $q_{\mathrm{NW}} = \lfloor 1.5\,|\mathcal{O}|^{1/3} \rfloor$ (its value on $\mathcal{T}_{\mathrm{opt}}$: Section 7.2).
- **Model confidence set** (Hansen et al., 2011), **superior predictive ability** (Hansen, 2005) and **StepM** (Romano and Wolf, 2005) for the comparison of predictor sets.

E14. Decision: drop the sentence, or report the set.
Problem: the model confidence set, the superior predictive ability test, and StepM are named, and the paper reports none of their outputs.
This bullet names three procedures and says they are for the comparison of predictor sets: the model confidence set, which returns the set of models not beaten at a stated confidence level; the superior predictive ability test, which returns a p-value against a benchmark; and StepM, which steps through the models and rejects some of them. No table in the paper states an MCS membership, an SPA p-value, or a StepM rejection. The next bullet says the main tables are point estimates and contain no bootstrap intervals, which is the kind of output those three procedures produce. The comparisons that are printed are Diebold–Mariano statistics on QLIKE and on squared error.

- **Point estimates only;** no bootstrap confidence intervals in the main tables.

I Notes to the authors (not part of the paper)

- **The results text anchors on the pooled block ridge;** re-base it on the per-bar OLS. Against that OLS, 7 of 111 forecasts have lower QLIKE, none significantly.
- **Data issue to fix before final numbers:**
  - the activity flags are computed after the forward fill, so the FOMC flags read 1 on all 122 trade days after November 2023;
  - VIX, volume-demand and FOMC feeds end early and are carried forward without limit into 2024.
- **Missing outputs:** paired-comparison tables (axis 1, axis 2), DM against the OLS in the master table, and the model confidence set over models.
- **Pending:** the pooled random forest (full table).
- **E15, sample and window still open** (2026-10-16). Hyperparameters: one configuration per model, chosen once on the pre-sample and frozen on the evaluation span. The daily HAR OLS stays outside the scheme. Not applied to Table 1.
  - Open choice: all 48 intervals, overnight included, or the 13 regular-hours bars (end stamps 10:00–16:00). Window: 24,000 intervals or 96,000 intervals (2,000 trading days on the 48-bar grid).
  - Calendar (note E9): columns after the overnight indicator are out of the specification, including the products of the target HAR components with the opening- and closing-interval indicators. The overnight indicator stays only if the overnight intervals stay in the training sample. On the 13 regular-hours bars it equals 1 only on the 16:00 bar. Not applied to the printed column counts or Table 1.
  - Shorter window, 16:00 rows only (closing-interval window study, 2026-10-03; trees refit every 10 sessions and the linear penalties are re-chosen, so this is not the frozen rule). LightGBM QLIKE 0.1110 at 24,000/48 sessions and 0.1006 at 2,000 sessions (DM 2.71). Ridge 0.1194 and 0.1027 (DM 2.94). Lasso 0.0998 and 0.1006 (DM −0.36).
  - A 48-bar pool against a one-bar fit, HAR plus calendar, 2012–2024. The pool preferred the shorter window; the one-bar fits preferred 2,000 days. At the 16:00 bar the one-bar QLIKE gain was 6–8% on the shorter window and 12–13% at 2,000 days. The full predictor set on one bar was worse on the shorter window (+4.1%, 8 bars worse) and flat at 2,000 days (+0.5%).
  - The 13 regular-hours bars as one model against the 48-bar pool, 2,000 days, mean over buckets. QLIKE lower by 1.75% / 1.86% / 1.85% (ridge / elastic net / lasso); at the close, 5.5–6.0%. The 15:30 sign trade, crossed, matched the pool (0.70 / 0.62 / 0.68 against the pool's 0.72 / 0.68 / 0.73). It moved only for a fit on the 16:00 bar alone

(QLIKE 8.6–9.5% below the pool; crossed Sharpe 1.00 / 0.92 / 0.85, lasso / elastic net / ridge).

- **The full outline,** with source and macro pointers, is `writeup/PAPER_REHAUL_OUTLINE_2026-10-15.md`.